PENGOLAHAN CITRA & VISI KOMPUTER

Segmentasi

(sumber: R.C. Gonzalez & R.E. Woods, Digital Image Processing, 4th edition)

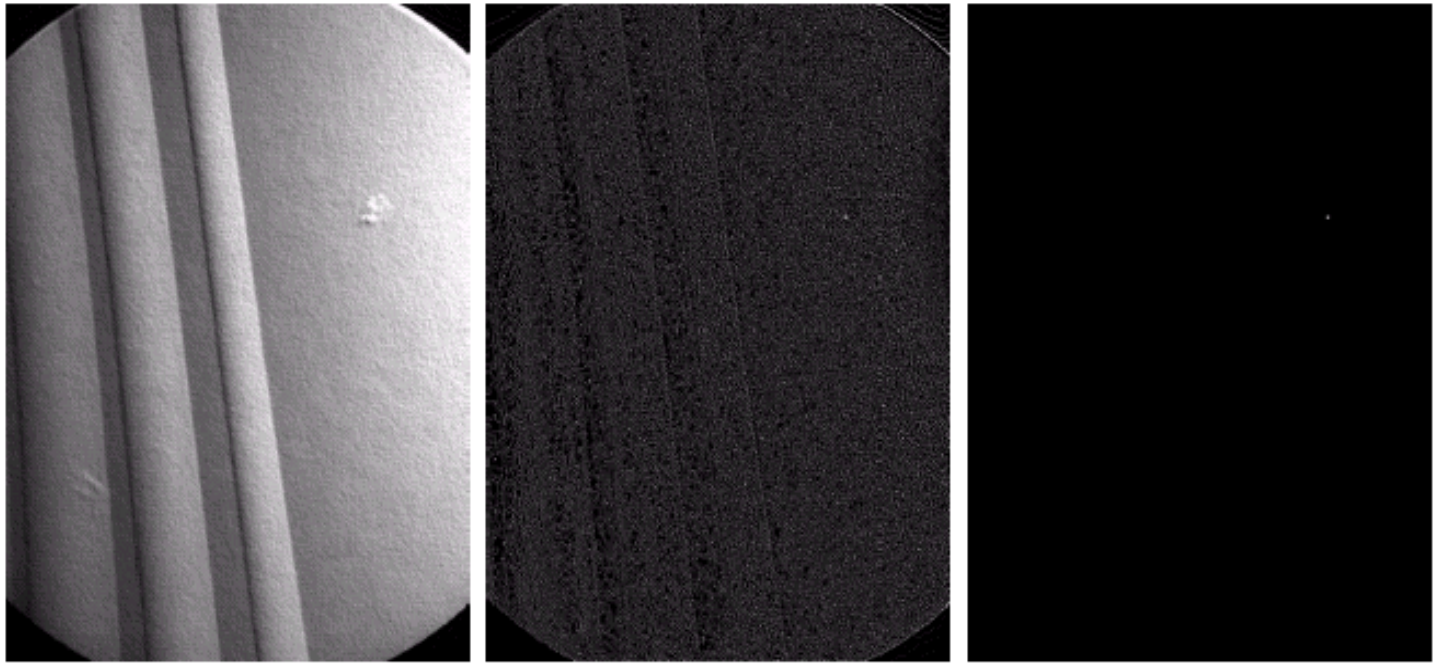
Handayani Tjandrasa

Segmentasi Citra

- Membagi citra dalam daerah2 dan objek2, proses segmentasi berhenti bila daerah/objek yang dikehendaki sudah terisolasi
- Algoritma berdasar:
  - Diskontinuitas (deteksi Titik, Garis, Tepi)
  - Similaritas

Deteksi Titik

−1	−1	−1
−1	8	−1
−1	−1	−1



a
b c d

FIGURE 10.2
(a) Point detection mask. (b) X-ray image of a turbine blade with a porosity. (c) Result of point detection. (d) Result of using Eq. (10.1-2). (Original image courtesy of X-TEK Systems Ltd.)

Deteksi Garis

−1	−1	−1
2	2	2
−1	−1	−1

Horizontal

−1	−1	2
−1	2	−1
2	−1	−1

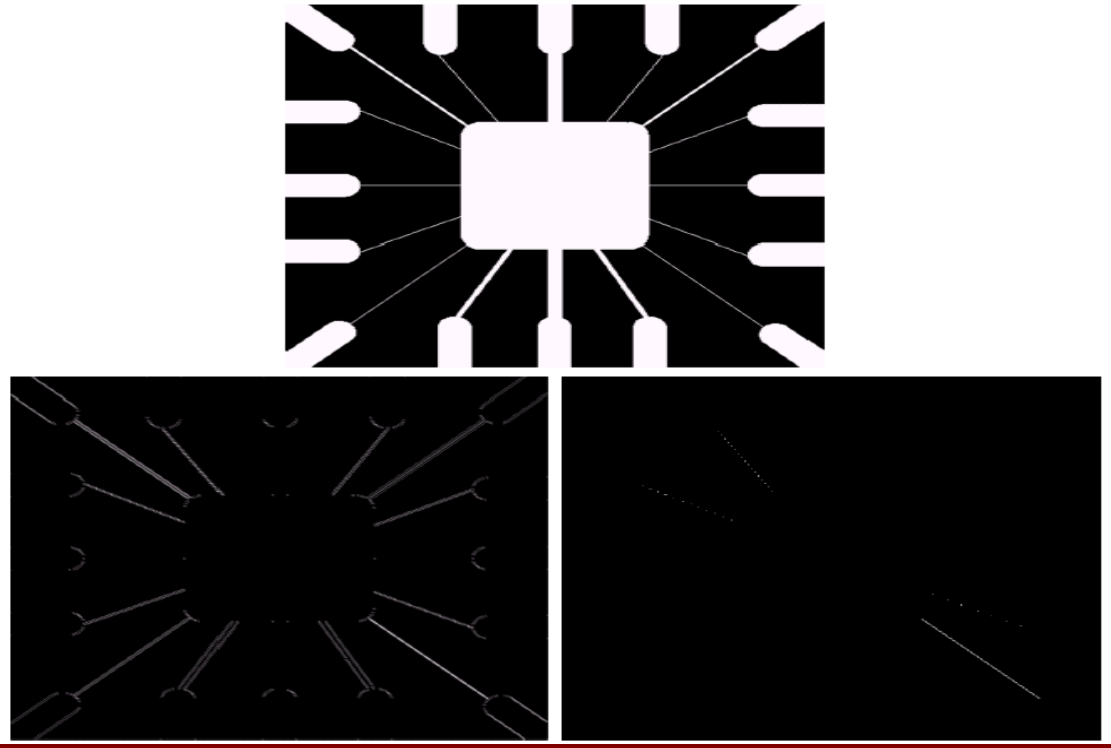
+45°

−1	2	−1
−1	2	−1
−1	2	−1

Vertical

2	−1	−1
−1	2	−1
−1	−1	2

−45°



a
b c

FIGURE 10.4 Illustration of line detection. (a) Binary wire-bond mask. (b) Absolute value of result after processing with −45° line detector. (c) Result of thresholding image (b).

Image Gradient

$$\nabla f(x,y) \equiv \text{grad}\left[f(x,y)\right] \equiv \begin{bmatrix} g_x(x,y) \\ g_y(x,y) \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f(x,y)}{\partial x} \\ \dfrac{\partial f(x,y)}{\partial y} \end{bmatrix}$$

Magnitude:
Euclidean vector norm

$$M(x,y) = \left\|\nabla f(x,y)\right\| = \sqrt{g_x^2(x,y) + g_y^2(x,y)}$$

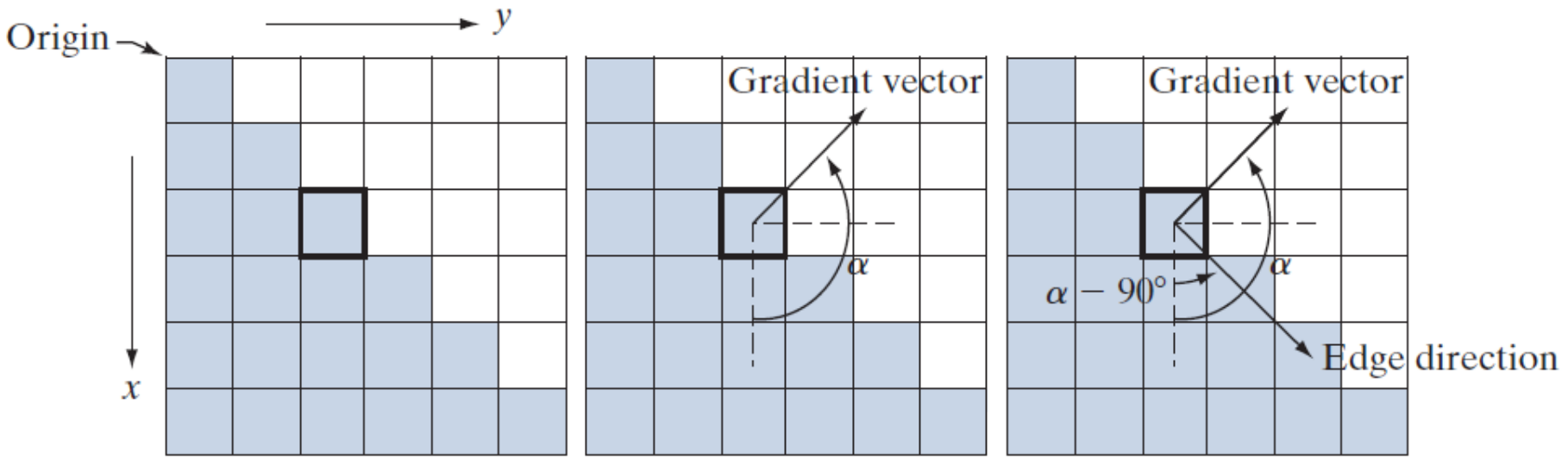
Absolute values

$$M(x,y) \approx \left|g_x\right| + \left|g_y\right|$$

Gradient vector direction

$$\alpha(x,y) = \tan^{-1}\left[\frac{g_y(x,y)}{g_x(x,y)}\right]$$

Vektor Gradient



a b c

FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge direction is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square represents one pixel. (Recall from Fig. 2.19 that the origin of our coordinate system is at the top, left.)

Operator Gradient

Pendekatan *forward difference*

$$g_x(x,y) = \frac{\partial f(x,y)}{\partial x} = f(x+1,y) - f(x,y)$$

$$g_y(x,y) = \frac{\partial f(x,y)}{\partial y} = f(x,y+1) - f(x,y)$$

−1
1

−1	1

Operator Gradient

Nilai intensitas z_i dari region citra 3x3

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Operator Roberts berbasis pada *diagonal difference*

$$g_x = \frac{\partial f}{\partial x} = (z_9 - z_5)$$

$$g_y = \frac{\partial f}{\partial y} = (z_8 - z_6)$$

Operator Gradient

Nilai intensitas z_i dari region citra 3x3

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Operator Prewitt berbasis dengan derivatif parsial

$$g_x = \frac{\partial f}{\partial x} = (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)$$

$$g_y = \frac{\partial f}{\partial y} = (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)$$

Operator Gradient

Nilai intensitas z dari region citra 3x3

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Operator Sobel berbasis dengan koefisien 2 ditengah memberikan hasil lebih baik

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$g_y = \frac{\partial f}{\partial y} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

Filter Gradient

FIGURE 10.14
A 3×3 region of an image (the z's are intensity values), and various kernels used to compute the gradient at the point labeled z_5.

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0	0	−1
0	1	1	0

Roberts

−1	−1	−1	−1	0	1
0	0	0	−1	0	1
1	1	1	−1	0	1

Prewitt

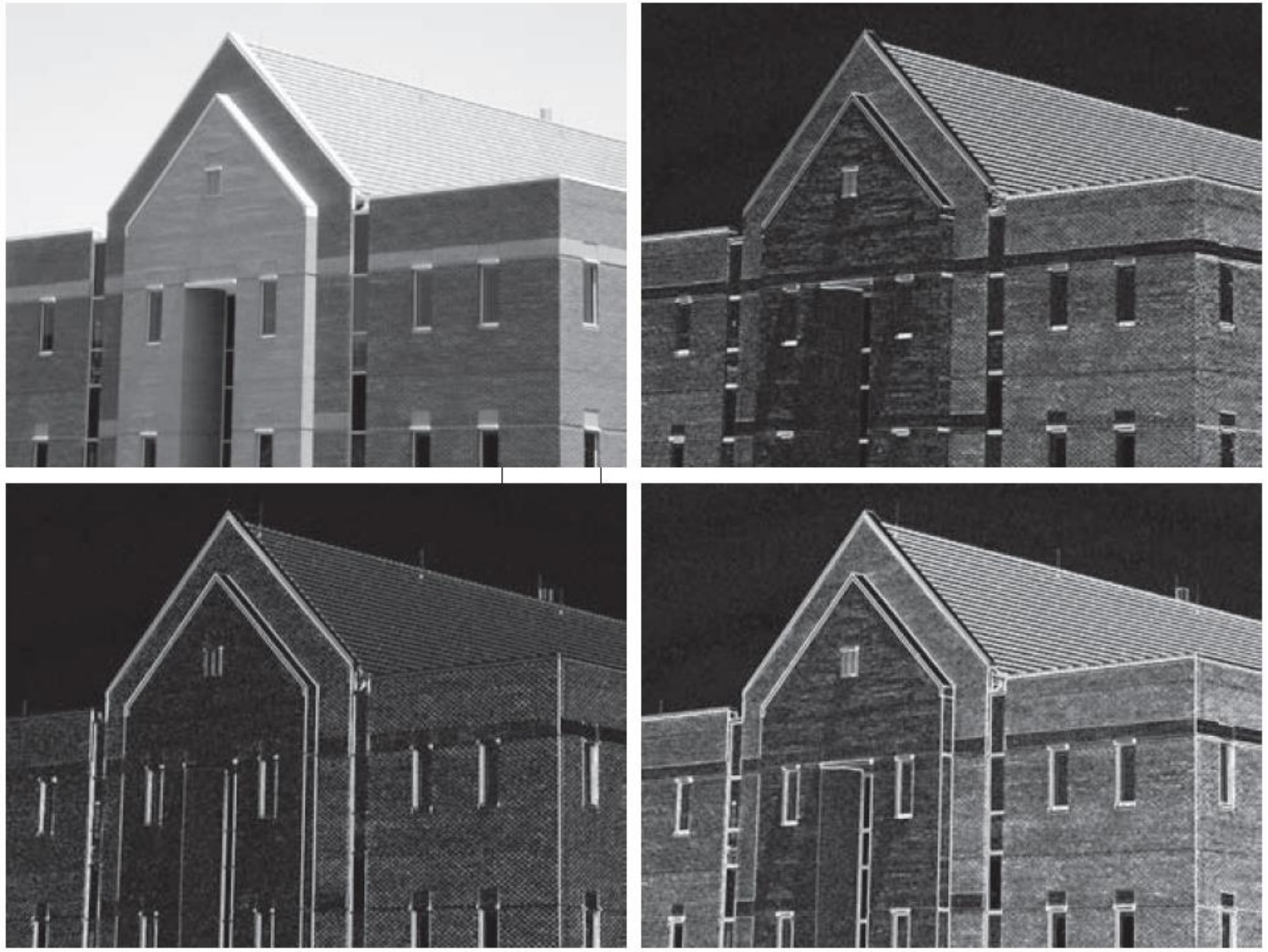
−1	−2	−1	−1	0	1
0	0	0	−2	0	2
1	2	1	−1	0	1

Sobel

Contoh Hasil Filter Gradient

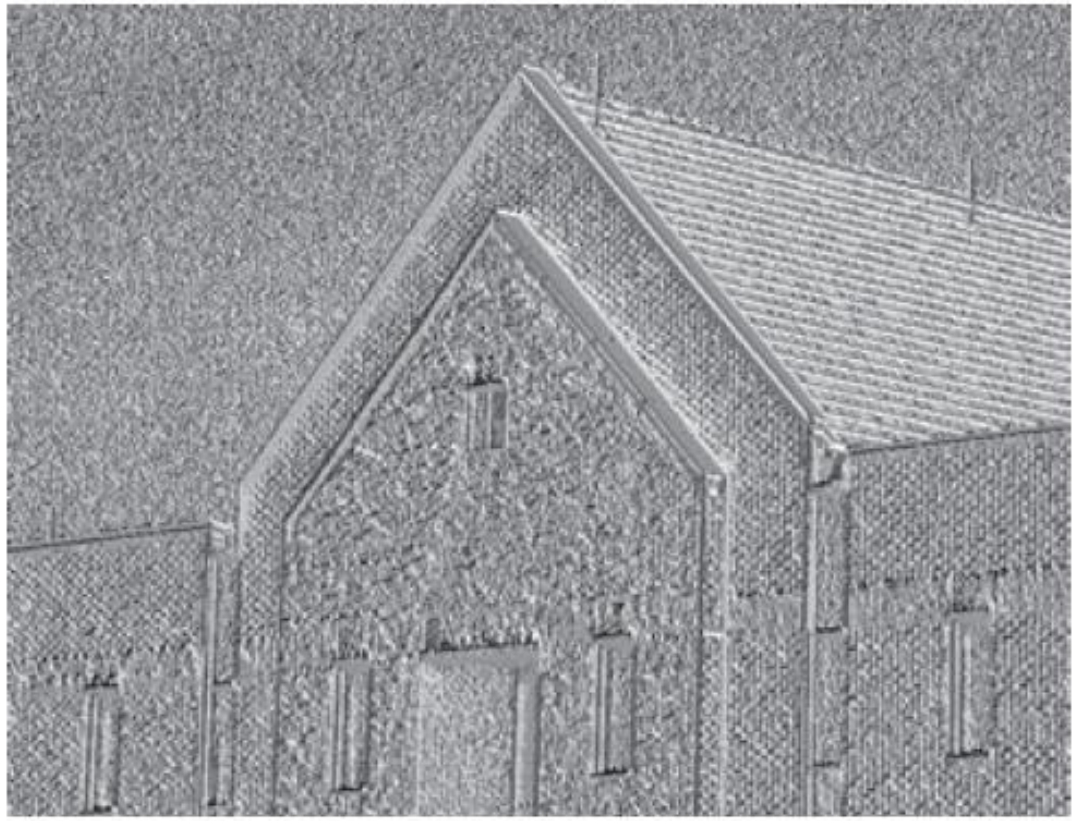
a b
c d

FIGURE 10.16
(a) Image of size 834×1114 pixels, with intensity values scaled to the range [0,1]. (b) $|g_x|$, the component of the gradient in the x-direction, obtained using the Sobel kernel in Fig. 10.14(f) to filter the image. (c) $|g_y|$, obtained using the kernel in Fig. 10.14(g). (d) The gradient image, $|g_x| + |g_y|$.



Gradient Angle

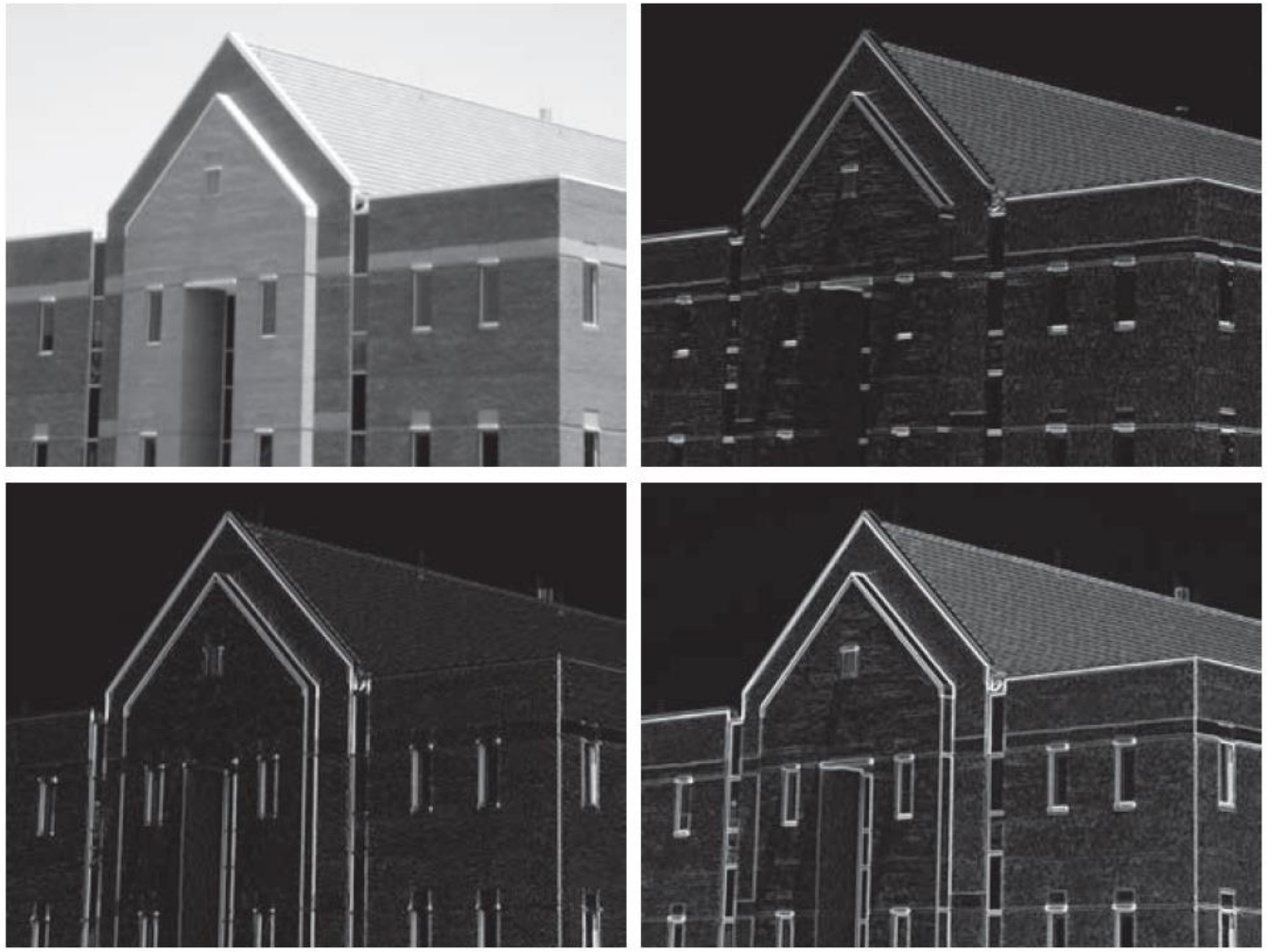
FIGURE 10.17
Gradient angle image computed using Eq. (10-18). Areas of constant intensity in this image indicate that the direction of the gradient vector is the same at all the pixel locations in those regions.



Contoh Hasil dengan Smoothing Sebelum Gradient

a b
c d

FIGURE 10.18
Same sequence as in Fig. 10.16, but with the original image smoothed using a 5×5 averaging kernel prior to edge detection.



Filter Laplacian

$$\nabla^2 f = \frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2}$$

$$\nabla^2 f = [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)]$$

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

Laplacian of Gaussian (LoG)

$$h(r) = -e^{-\frac{r^2}{2\sigma^2}}$$

Dimana $r^2 = x^2+y^2$, dan
σ Adalah standard deviasi

Laplacian dari h(r):

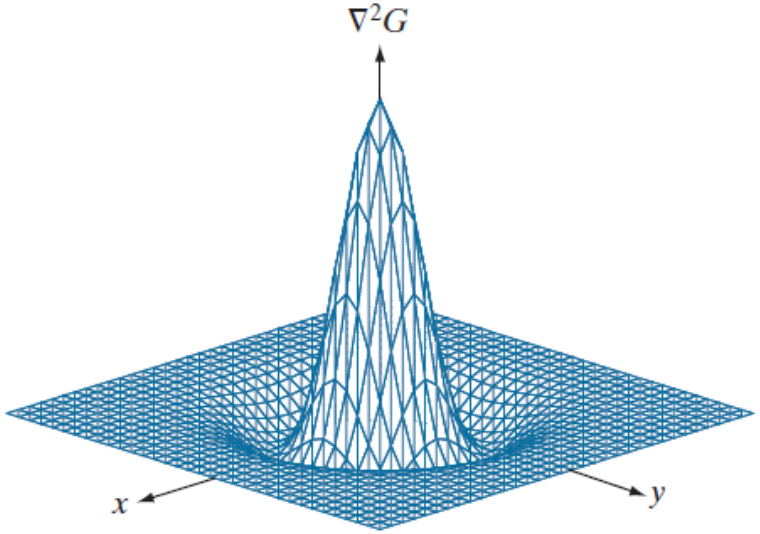
$$\nabla^2 h(r) = -\left[\frac{r^2 - \sigma^2}{\sigma^4}\right] e^{-\frac{r^2}{2\sigma^2}}$$

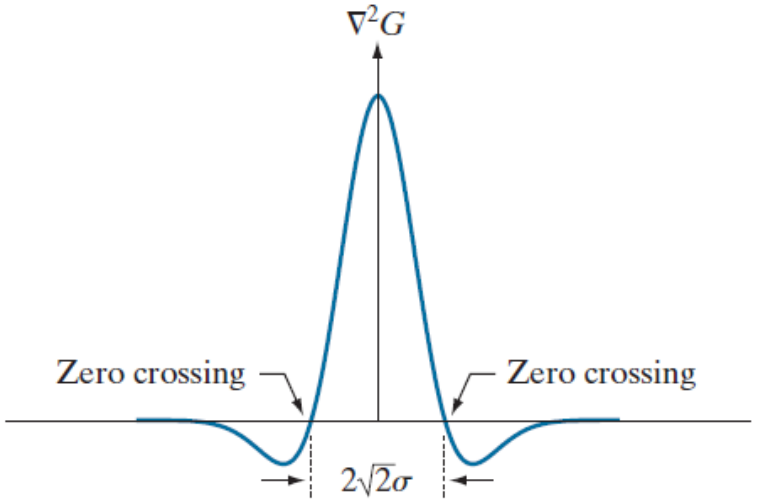
Gabungan Laplacian dan Gaussian untuk memperoleh tepi dengan zero crossing

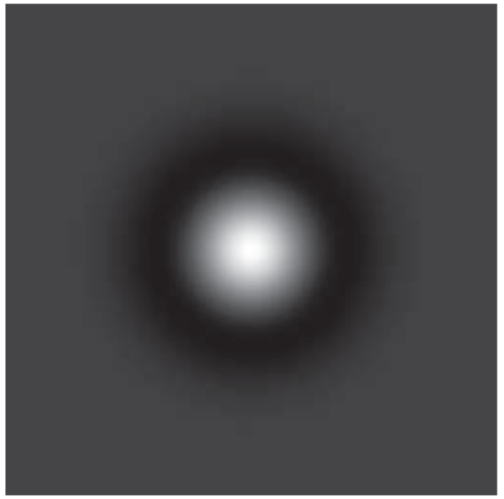
Filter berbentuk Topi Meksiko

a b
c d

FIGURE 10.21
(a) 3-D plot of the *negative* of the LoG.
(b) Negative of the LoG displayed as an image.
(c) Cross section of (a) showing zero crossings.
(d) 5×5 kernel approximation to the shape in (a). The negative of this kernel would be used in practice.

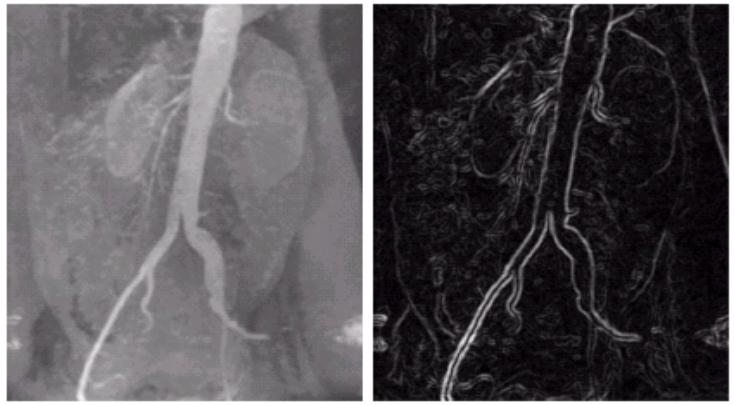


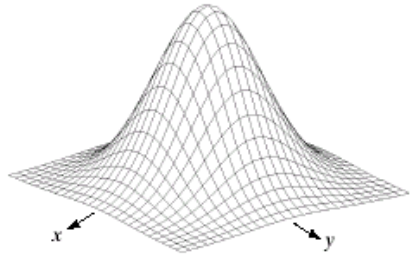




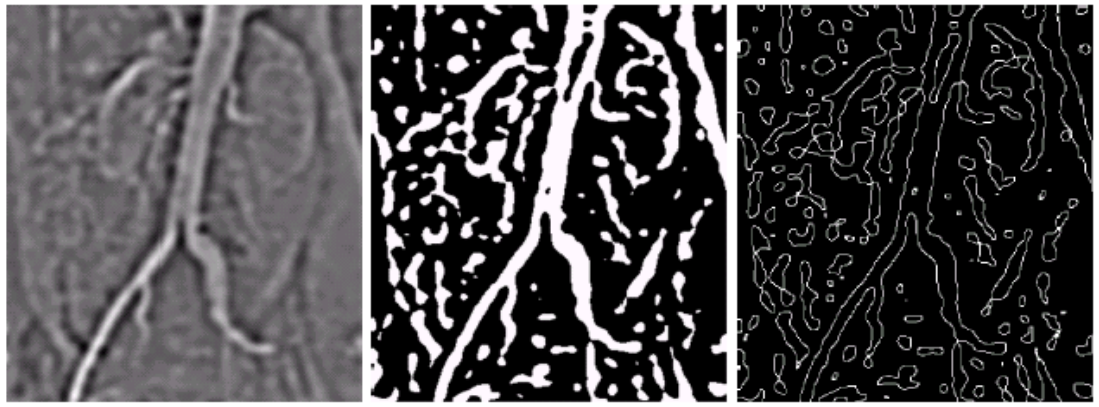
0	0	−1	0	0
0	−1	−2	−1	0
−1	−2	16	−2	−1
0	−1	−2	−1	0
0	0	−1	0	0

Contoh Hasil Gaussian&Laplacian





−1	−1	−1
−1	8	−1
−1	−1	−1



a. Citra awal
b. Gradient Sobel
c. Fungsi spasial smoothing Gaussian
d. Filter Laplacian
e. LoG
f. Threshold LoG
g. Zero crossing

Penyambungan tepi dengan kriteria gradient

Piksel dengan koordinate (x_0,y_0) mempunyai besaran (magnitude) yang sama dengan piksel pada (x,y) bila

$$|\nabla f(x,y) - \nabla f(x_0,y_0)| \leq E$$

Dan mempunyai sudut yang sama bila

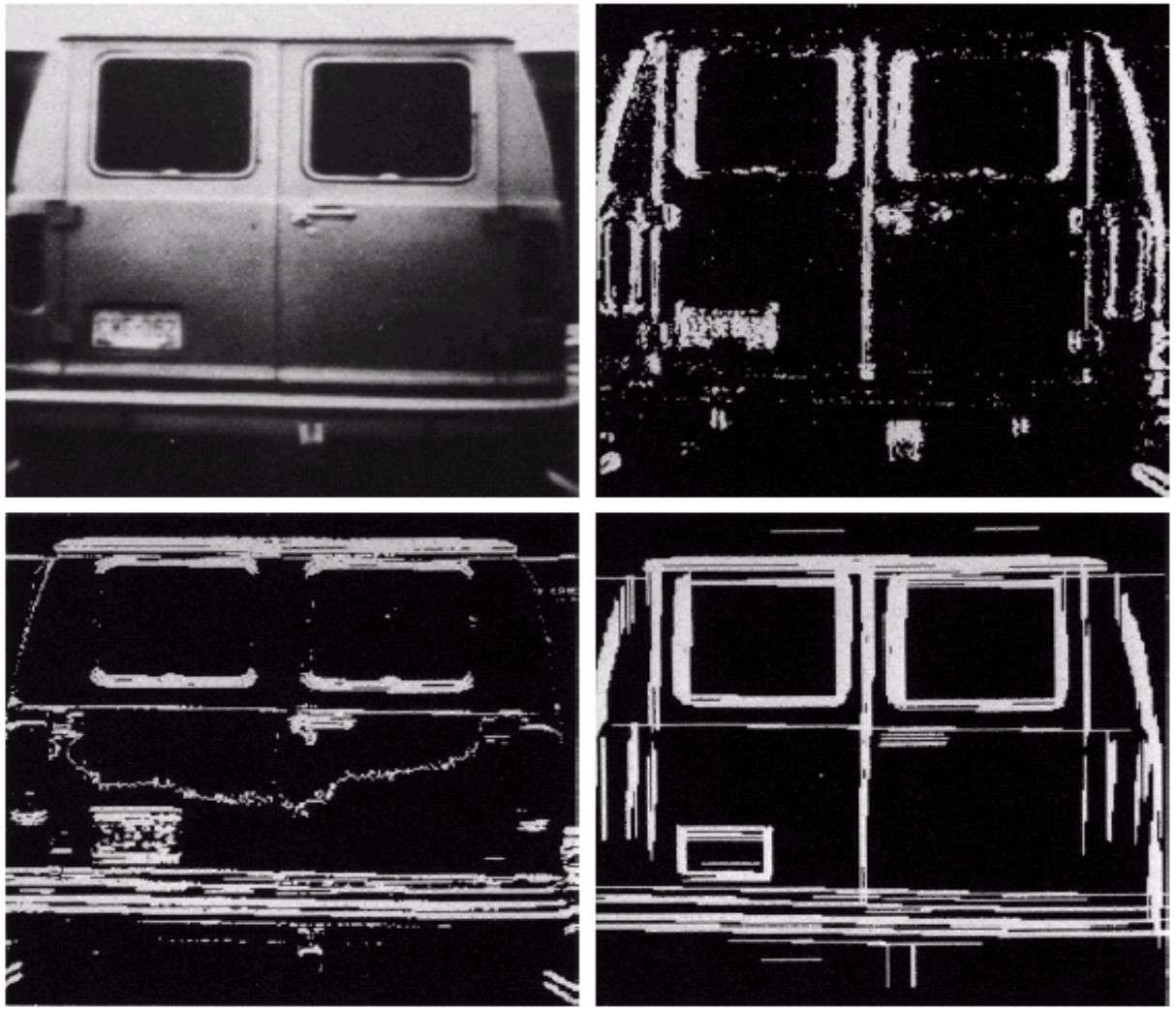
$$|\alpha(x,y) - \alpha(x_0,y_0)| < A$$

Titik terhubung ke tetangga pada (x,y) bila kedua kriteria diatas dipenuhi

Contoh proses citra untuk mendapatkan segi4 nomer plat mobil

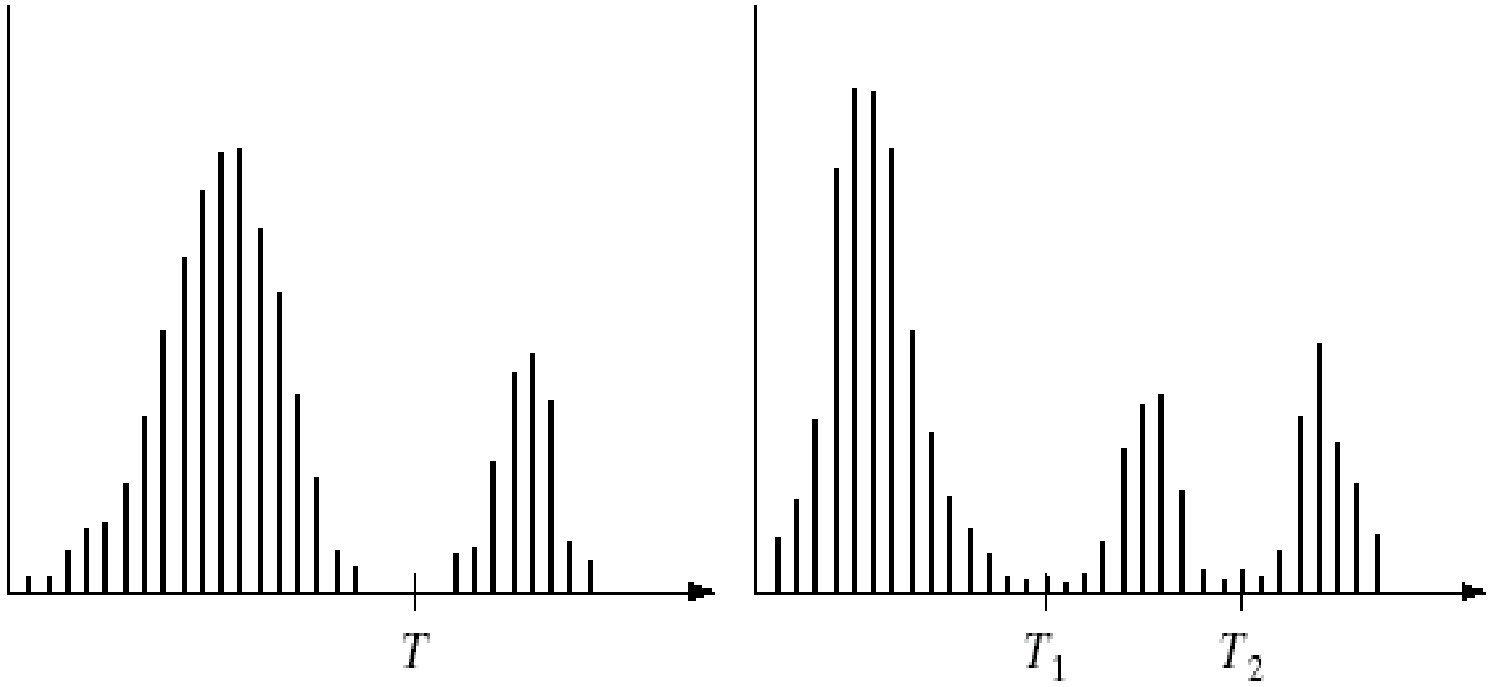
a b
c d

FIGURE 10.16
(a) Input image.
(b) G_y component of the gradient.
(c) G_x component of the gradient.
(d) Result of edge linking. (Courtesy of Perceptics Corporation.)



Digunakan operator Sobel horisontal dan vertikal dengan kondisi terhubung bila:
nilai gradient >25, dan perbedaan arah sudut <15°

Thresholding



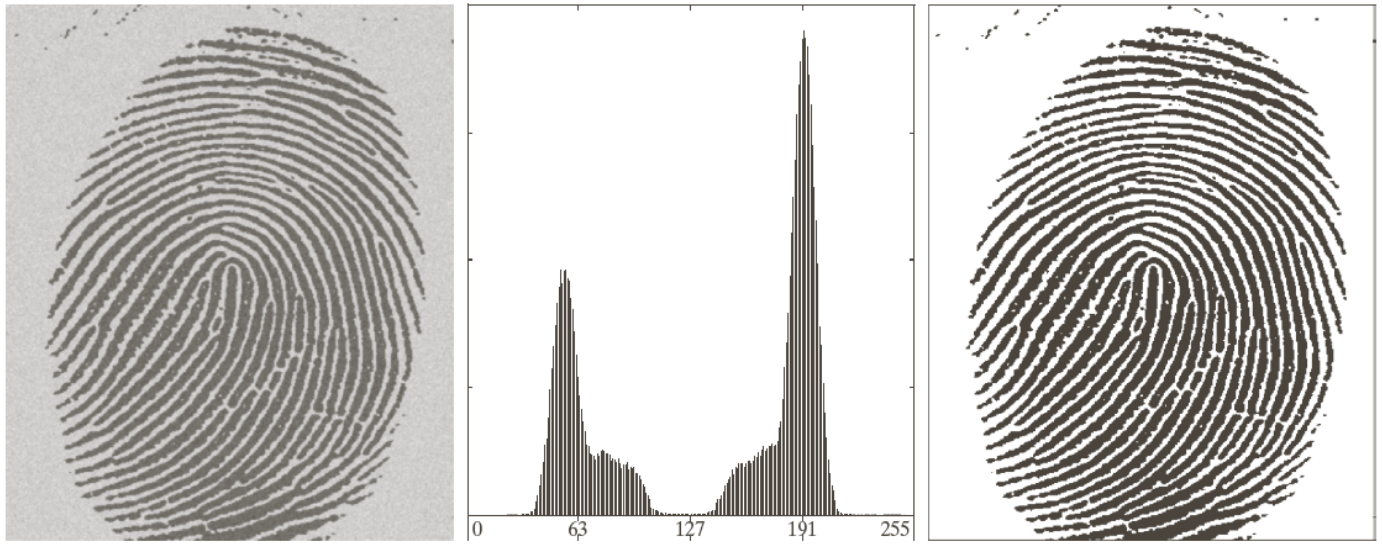
a b

FIGURE 10.26 (a) Gray-level histograms that can be partitioned by (a) a single threshold, and (b) multiple thresholds.

Basic Thresholding

1. Pilih threshold awal *T*
2. *T* membagi histogram menjadi 2 grup piksel *G1* dan *G2*
3. Hitung intensitas rata-rata grup *m1, m2*
4. *T=(m1+m2)/2*
5. Ulangi proses 2-4 sampai ΔT cukup kecil

Contoh Global Threshold



a b c

FIGURE 10.38 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (the border was added for clarity). (Original courtesy of the National Institute of Standards and Technology.)

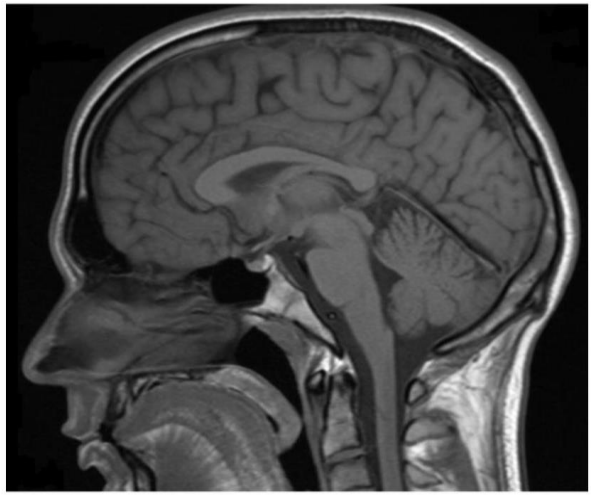
Metode Otsu

- Metode ini memaksimumkan varian antar kelas (between-class variance)
- Anggap nilai intensitas piksel 0 – (L-1)
- Dengan threshold $T(k)=k,\ 0 \le k \le L-1$ bagi citra menjadi 2 kelas C_1 dan C_2, dan membagi histogram ternormalisasi (=1) menjadi P_1 dan $P_2 = 1-P_1$ dengan intensitas rata-rata m_1 dan m_2
- Varian antar kelas: $\sigma^2_B = P_1(m_1 - m_G)^2 + P_2(m_2 - m_G)^2$
- Intensitas kumulatif rata-rata: $m(k) = m_1(k)\ P_1(k)$
- Nilai k diperoleh dengan memaksimalkan:

$$\sigma^2_B(k) = (m_G P_1(k) - m(k))^2 / (P_1(k)(1 - P_1(k)))$$

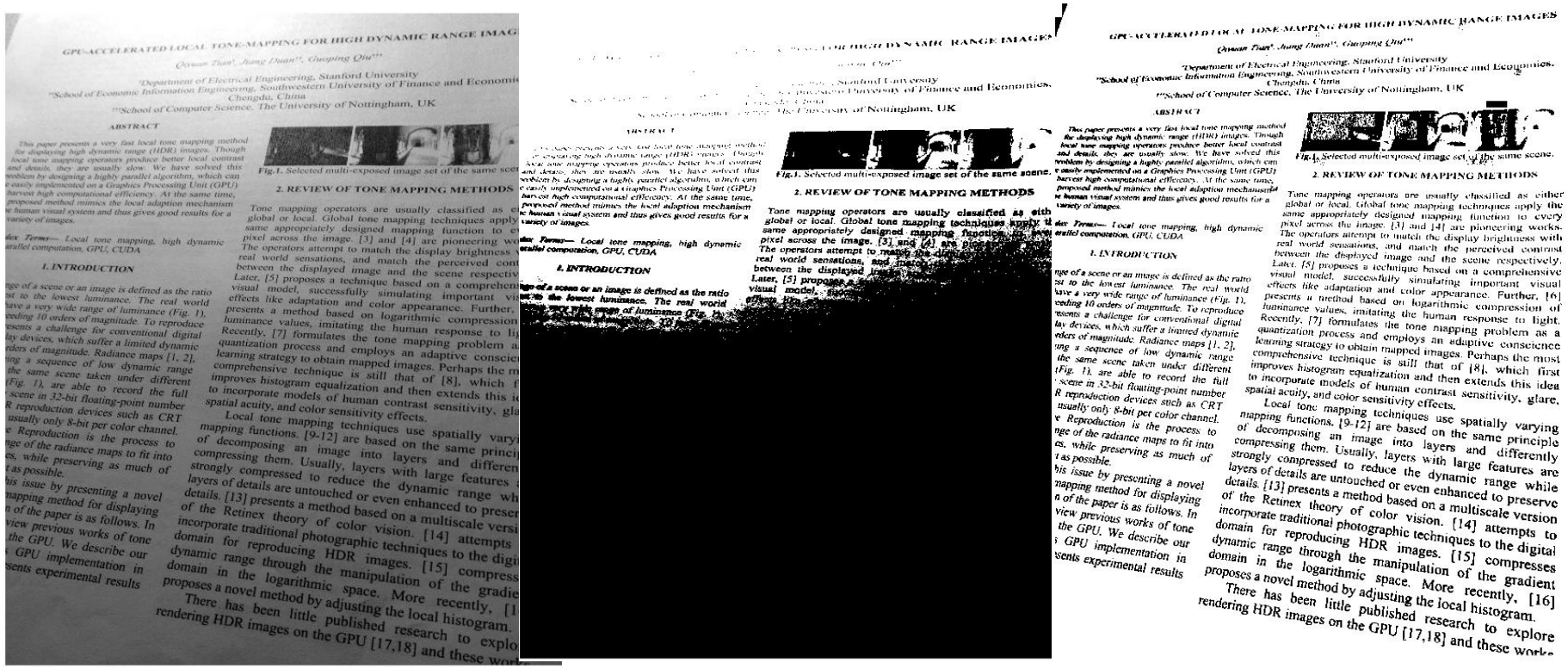
Program Matlab

```
clear all, close all;
% Read test image
Im = imread ('brain.jpg');
% Threshold value
thres = graythresh(Im);
% Get binary image
bwIm = im2bw(Im,thres);
figure ('Name', 'original image'), imshow(Im);
figure ('Name', 'global thresholded image'), imshow(bwIm);
```

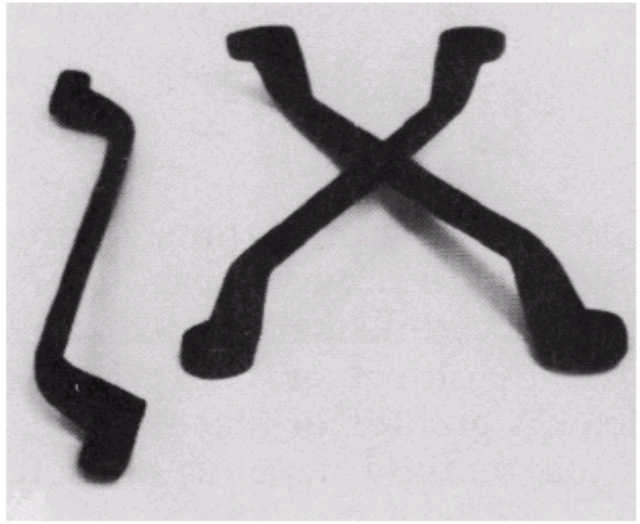


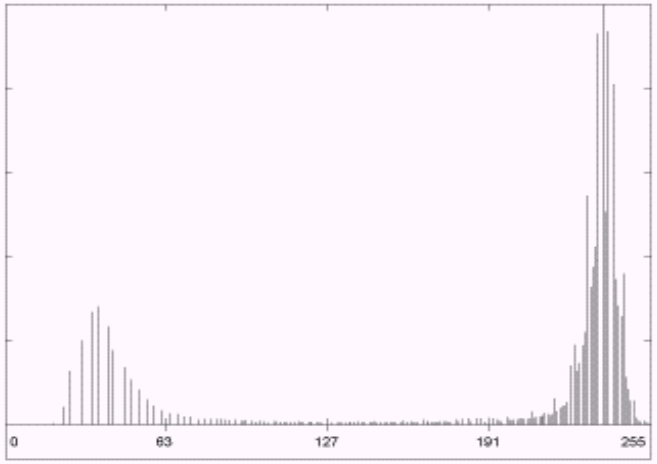


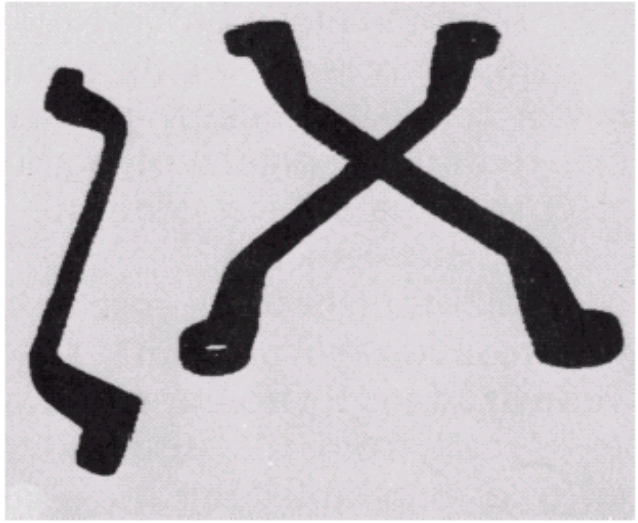
Hasil dengan Global Thresholding dan Locally Adaptive Thresholding



Global Thresholding



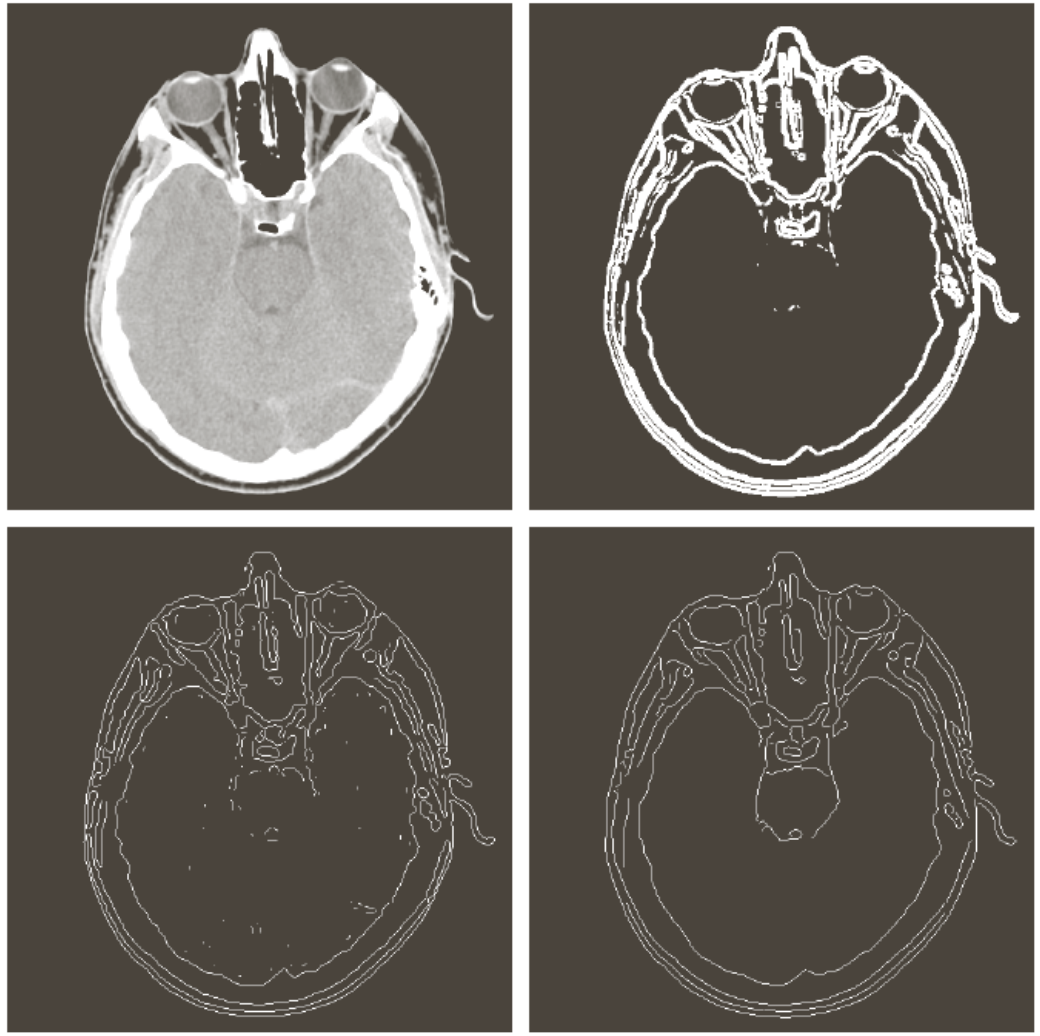




a
b c

FIGURE 10.28
(a) Original image. (b) Image histogram. (c) Result of global thresholding with T midway between the maximum and minimum gray levels.

Contoh Thresholded Gradient

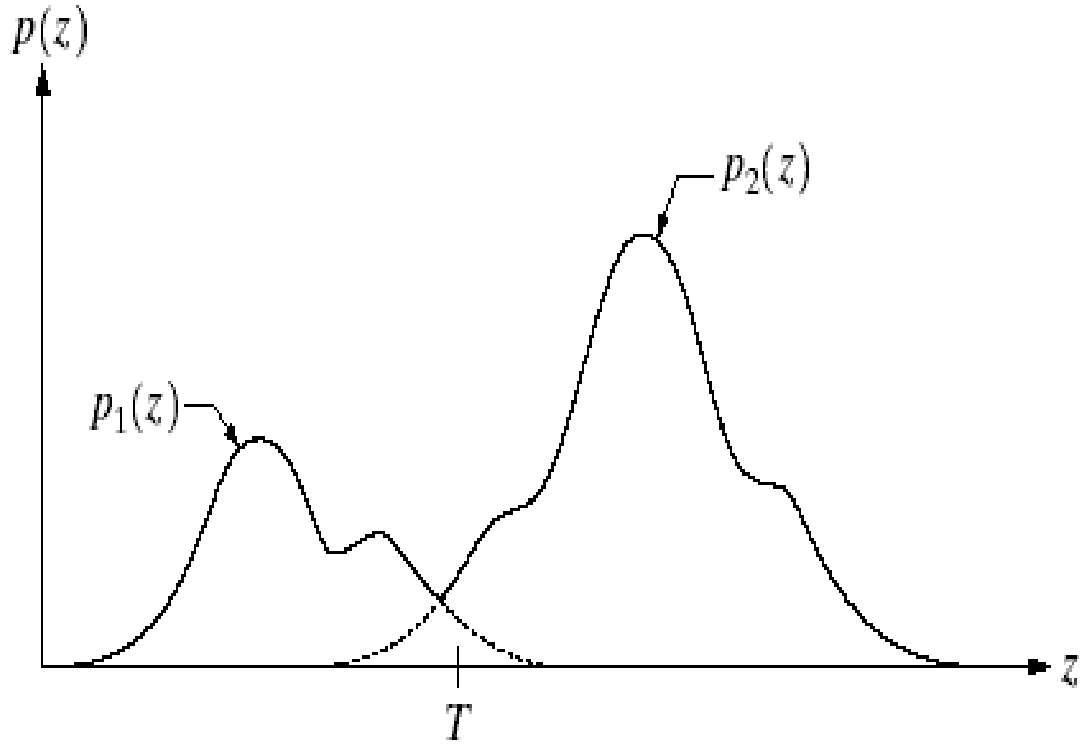


a b
c d

FIGURE 10.26
(a) Original head CT image of size 512 × 512 pixels, with intensity values scaled to the range [0, 1]. (b) Thresholded gradient of smoothed image. (c) Image obtained using the Marr-Hildreth algorithm. (d) Image obtained using the Canny algorithm. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

Optimal dan Adaptive Thresholding

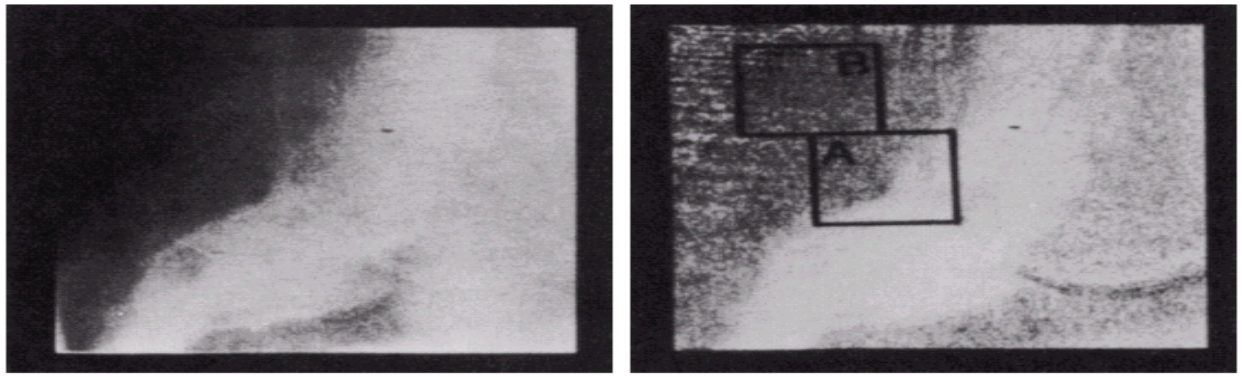
FIGURE 10.32
Gray-level probability density functions of two regions in an image.

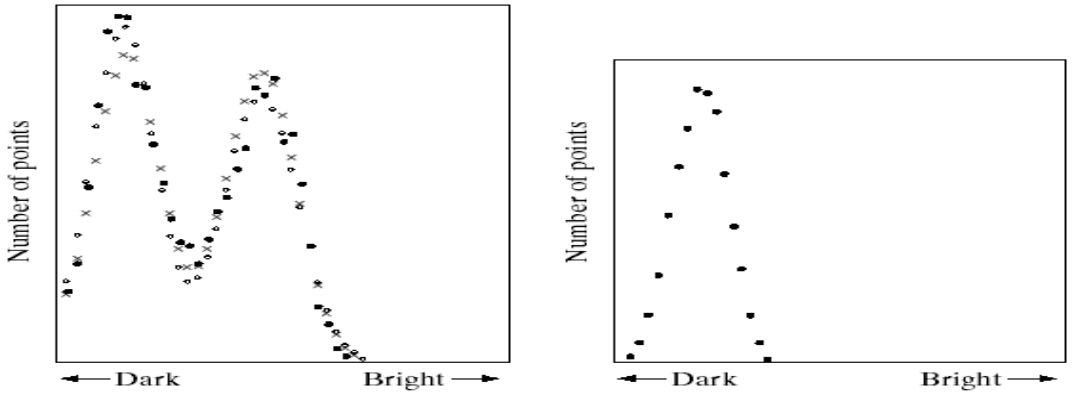


Contoh Adaptive Thresholding

a b

FIGURE 10.33 A cardioangiogram before and after preprocessing. (Chow and Kaneko.)





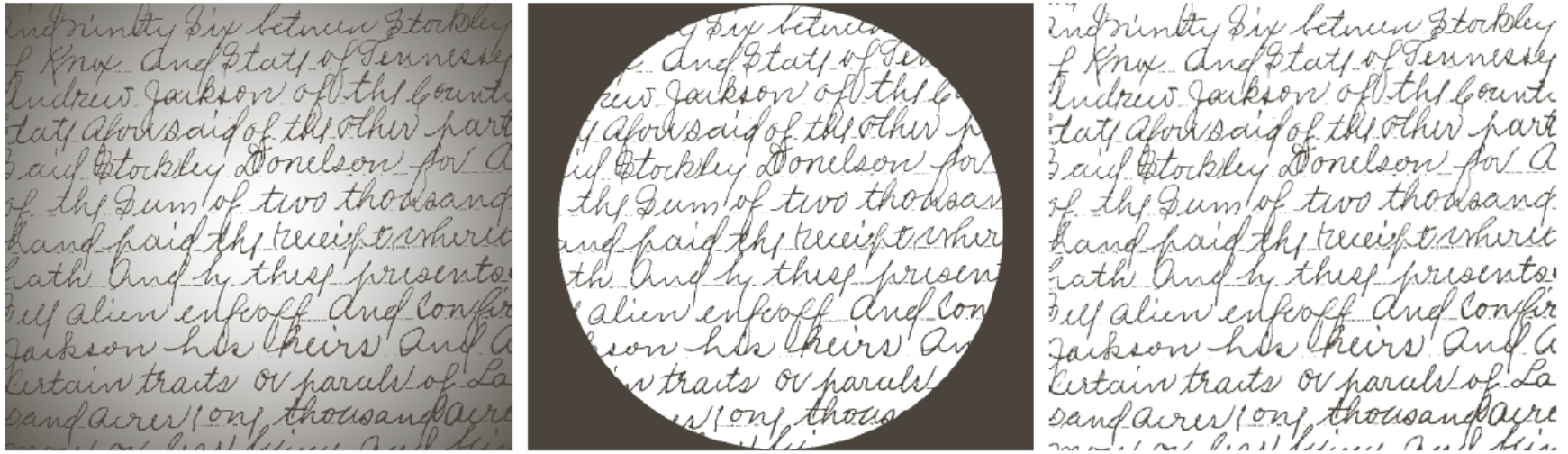
a b

FIGURE 10.34 Histograms (black dots) of (a) region *A*, and (b) region *B* in Fig. 10.33(b). (Chow and Kaneko.)

FIGURE 10.35 Cardioangiogram showing superimposed boundaries. (Chow and Kaneko.)



Segmentasi dengan iluminasi tidak rata

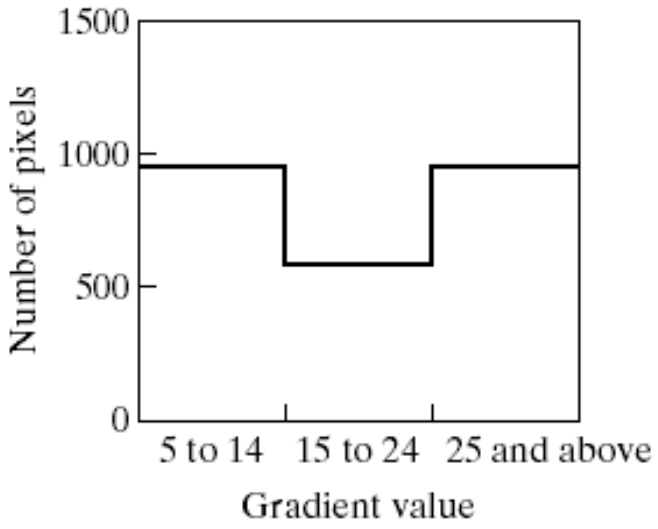


a b c

FIGURE 10.49 (a) Text image corrupted by spot shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.

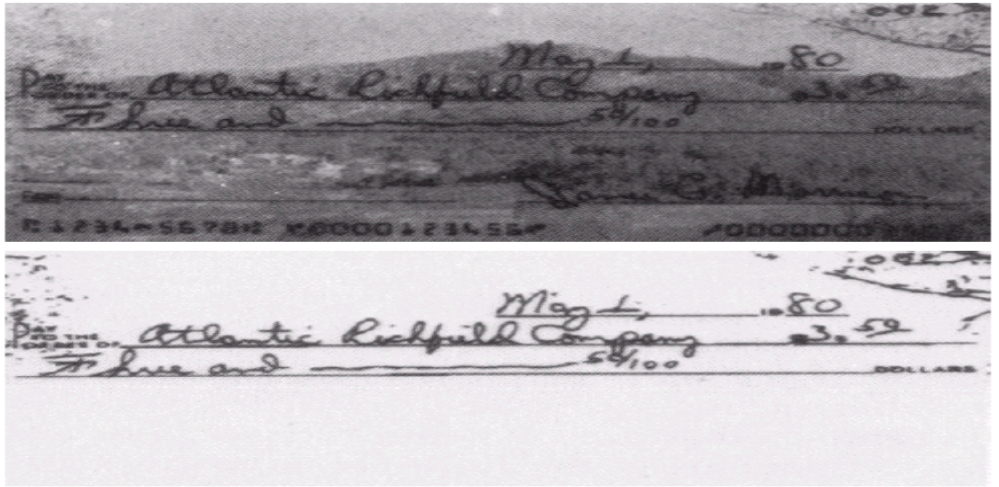
Contoh Histogram dari Gradient Citra

FIGURE 10.38
Histogram of pixels with gradients greater than 5. (Courtesy of IBM Corporation.)



a
b

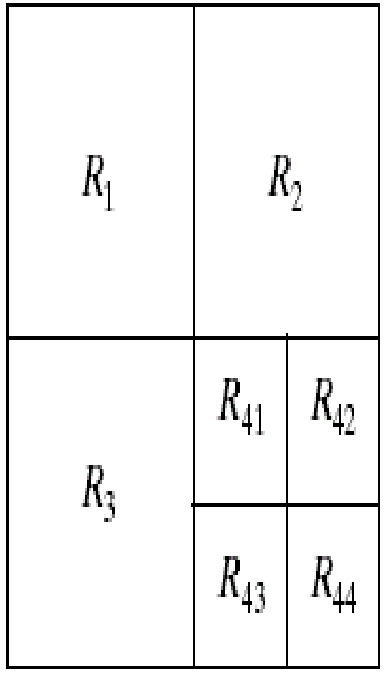
FIGURE 10.37
(a) Original image. (b) Image segmented by local thresholding. (Courtesy of IBM Corporation.)

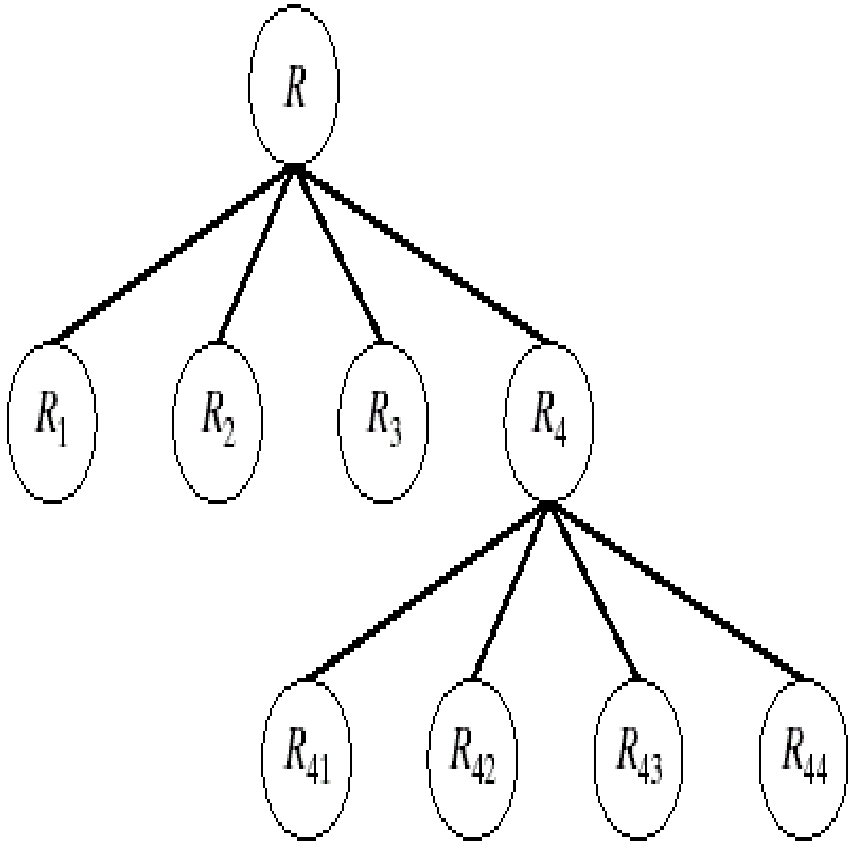


Region Splitting& Merging

a b

FIGURE 10.42
(a) Partitioned image.
(b) Corresponding quadtree.

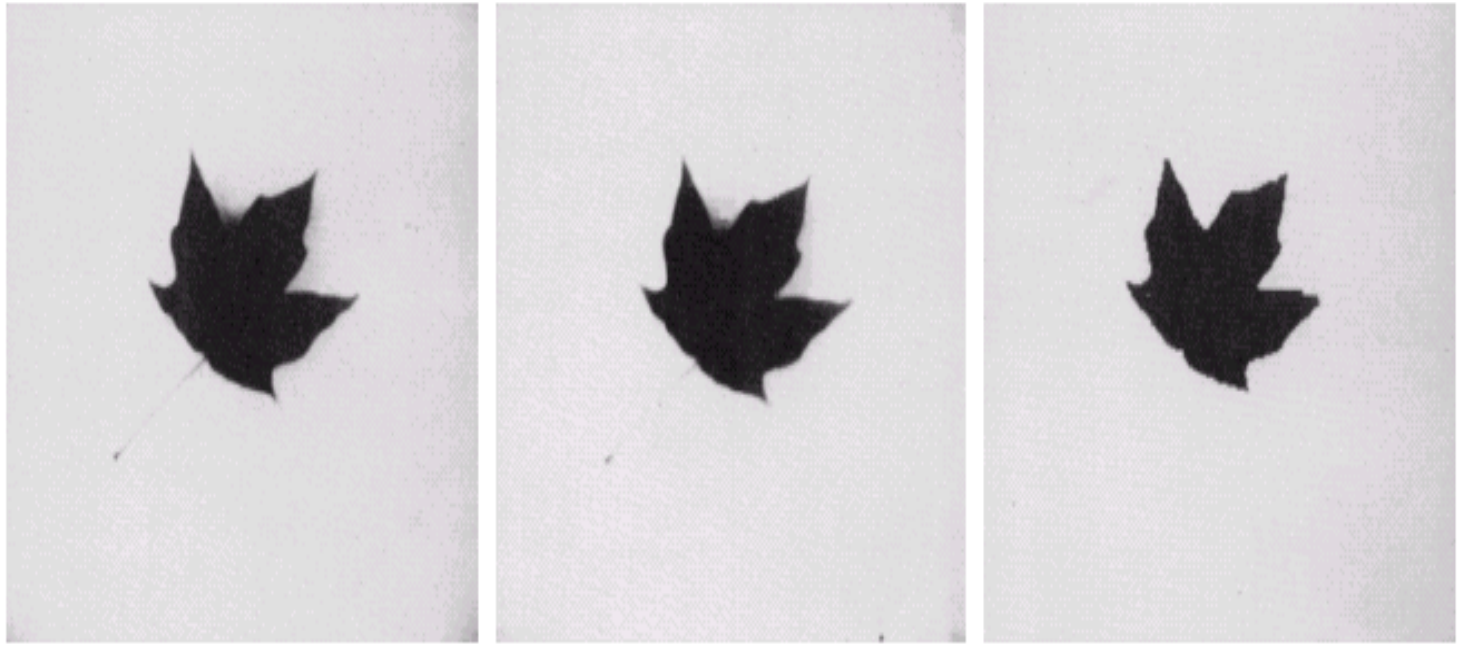




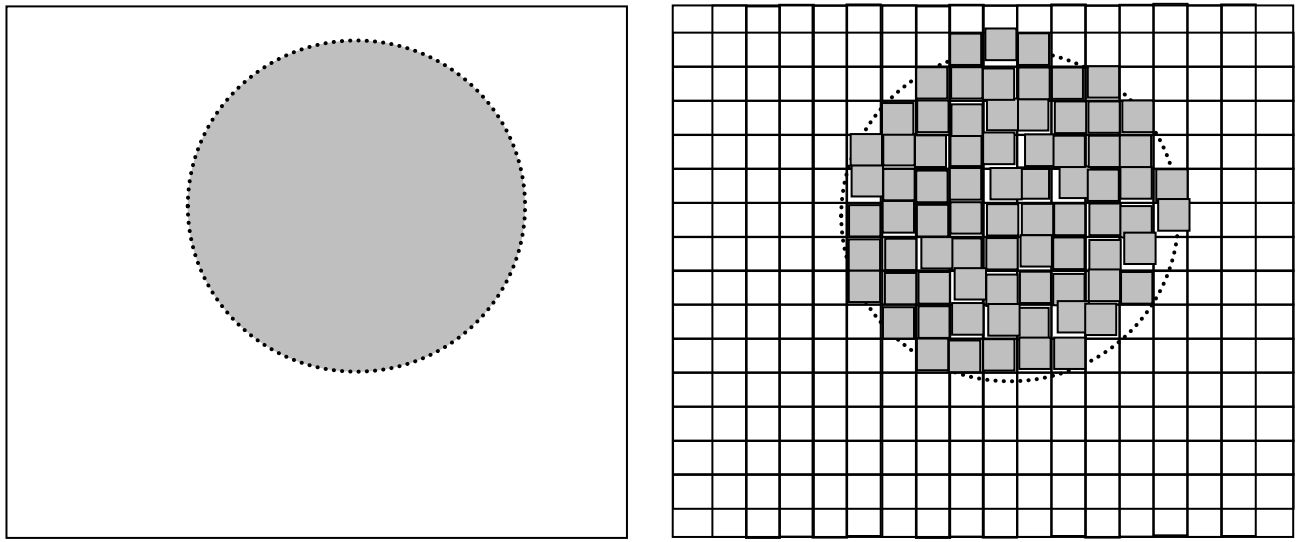
Contoh Hasil Splitting&Merging

a b c

FIGURE 10.43
(a) Original image. (b) Result of split and merge procedure. (c) Result of thresholding (a).



Region Growing



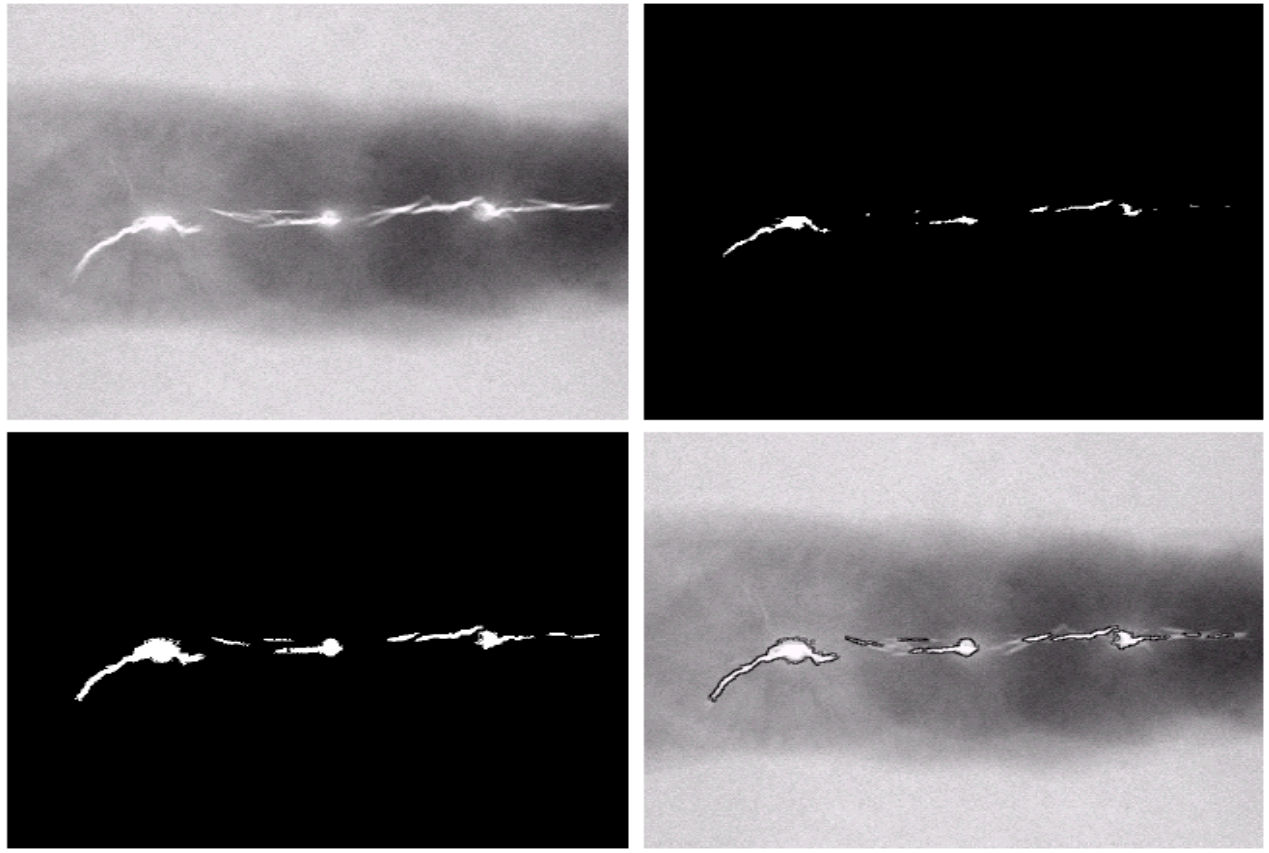
- Inisialisasi dengan *seed*
- Agregasi piksel tetangga yang selisih nilainya dengan *seed* tidak melebihi threshold
- Ulangi sampai tidak ada perubahan

Region Growing

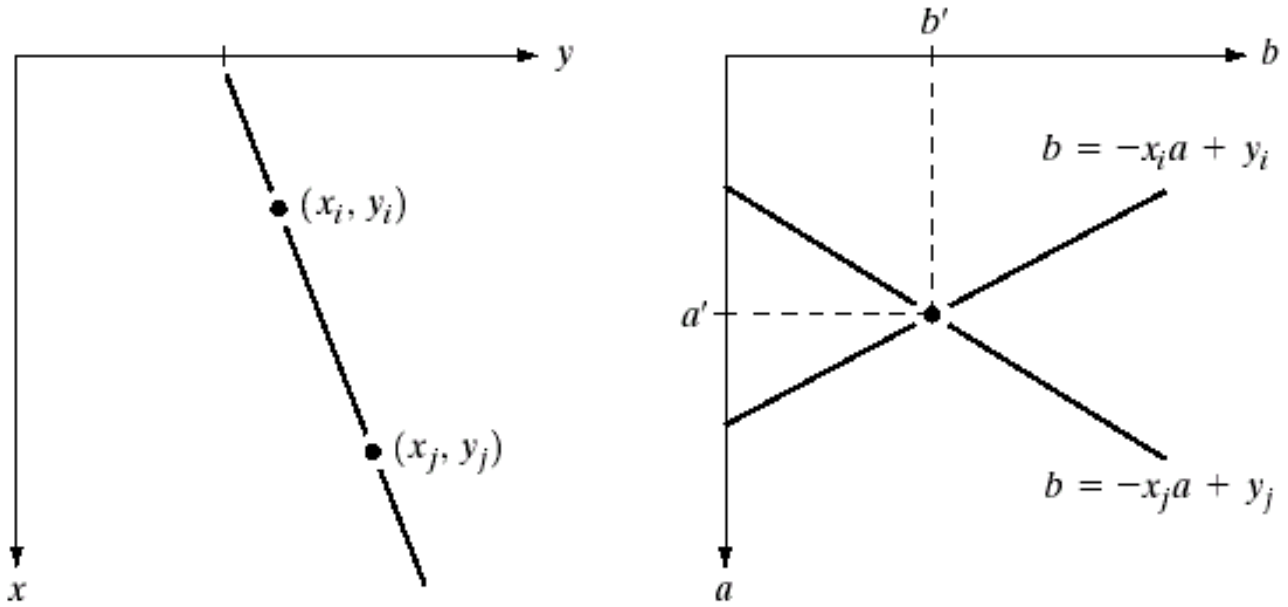
Mulai dari titik biji (seed) berkembang dengan menambahkan titik2 tetangga yang punya karakter sama

a b
c d

FIGURE 10.40
(a) Image showing defective welds. (b) Seed points. (c) Result of region growing. (d) Boundaries of segmented defective welds (in black). (Original image courtesy of X-TEK Systems, Ltd.).



Hough Transformation untuk garis



a b

FIGURE 10.17
(a) *xy*-plane. (b) Parameter space.

xy-plane

ab-plane or parameter space

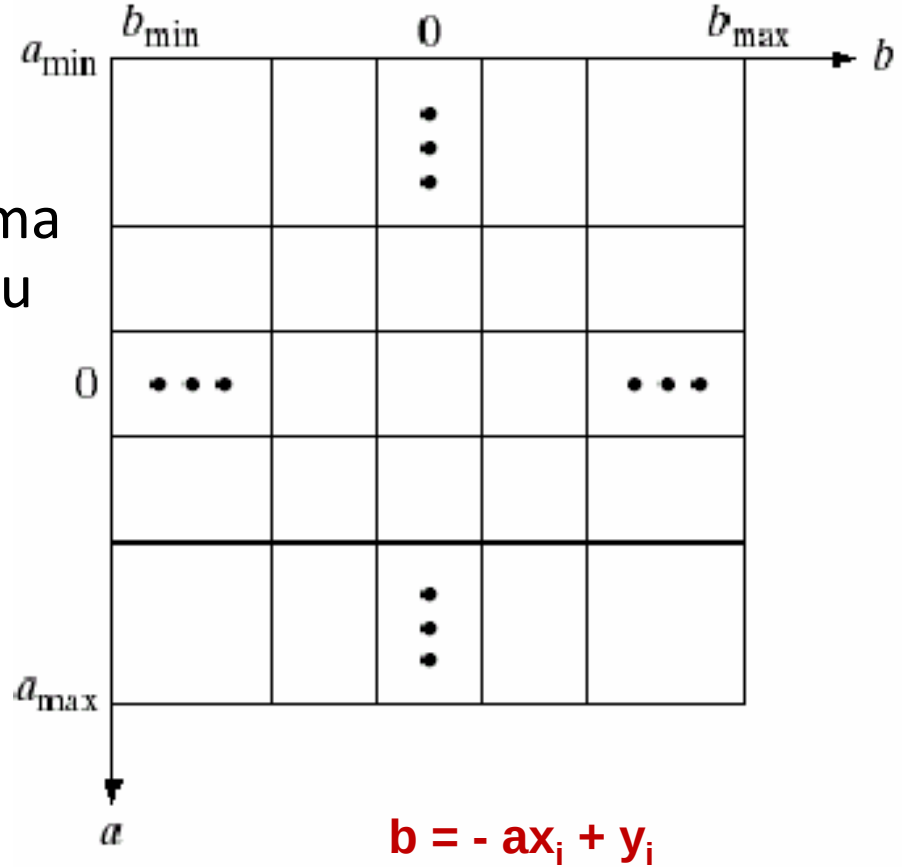
$y_i = ax_i + b$ ⟹ $b = - ax_i + y_i$

Titik-titik (x_i ,y_i) pada satu garis menghasilkan garis-garis yang berpotongan pada satu nilai (a',b') dalam ruang parameter.

Accumulator

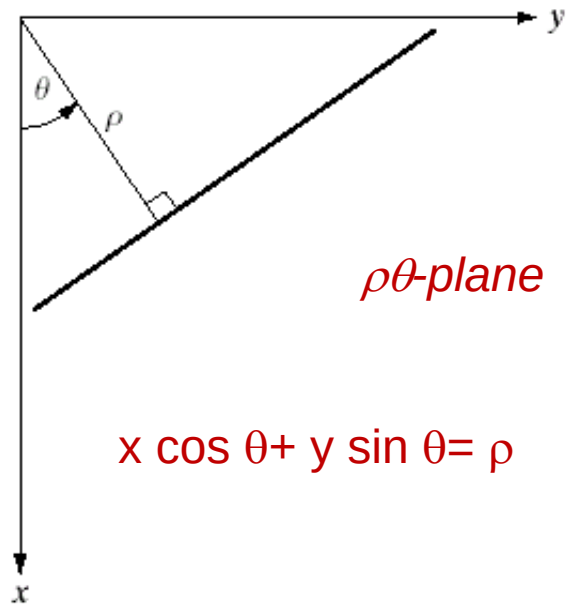
- Mula-mula akumulator diinisialisasi = 0
- Titik-titik pada garis yang sama menambah jumlah nilai suatu akumulator
- Batas nilai:

  (a_{max}, a_{min}) dan (b_{max}, b_{min})



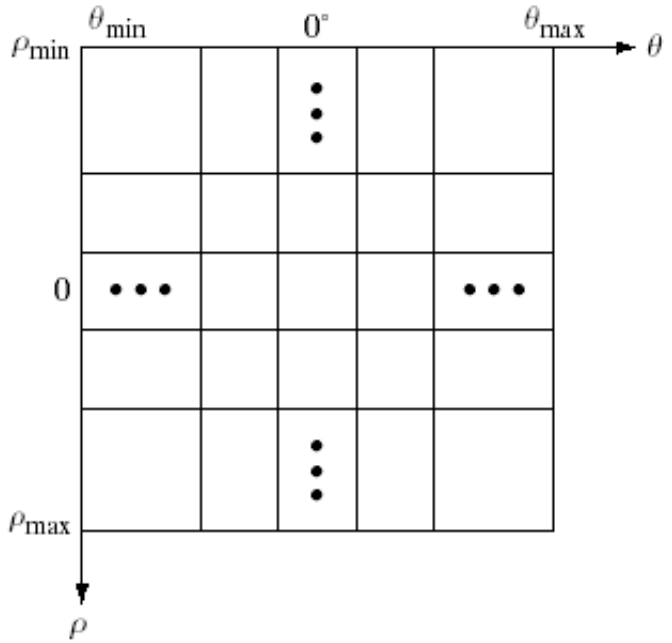
$b = - ax_i + y_i$

ρθ-plane



ρθ-plane

x cos θ+ y sin θ= ρ



a b

FIGURE 10.19
(a) Normal representation of a line.
(b) Subdivision of the ρθ-plane into cells.

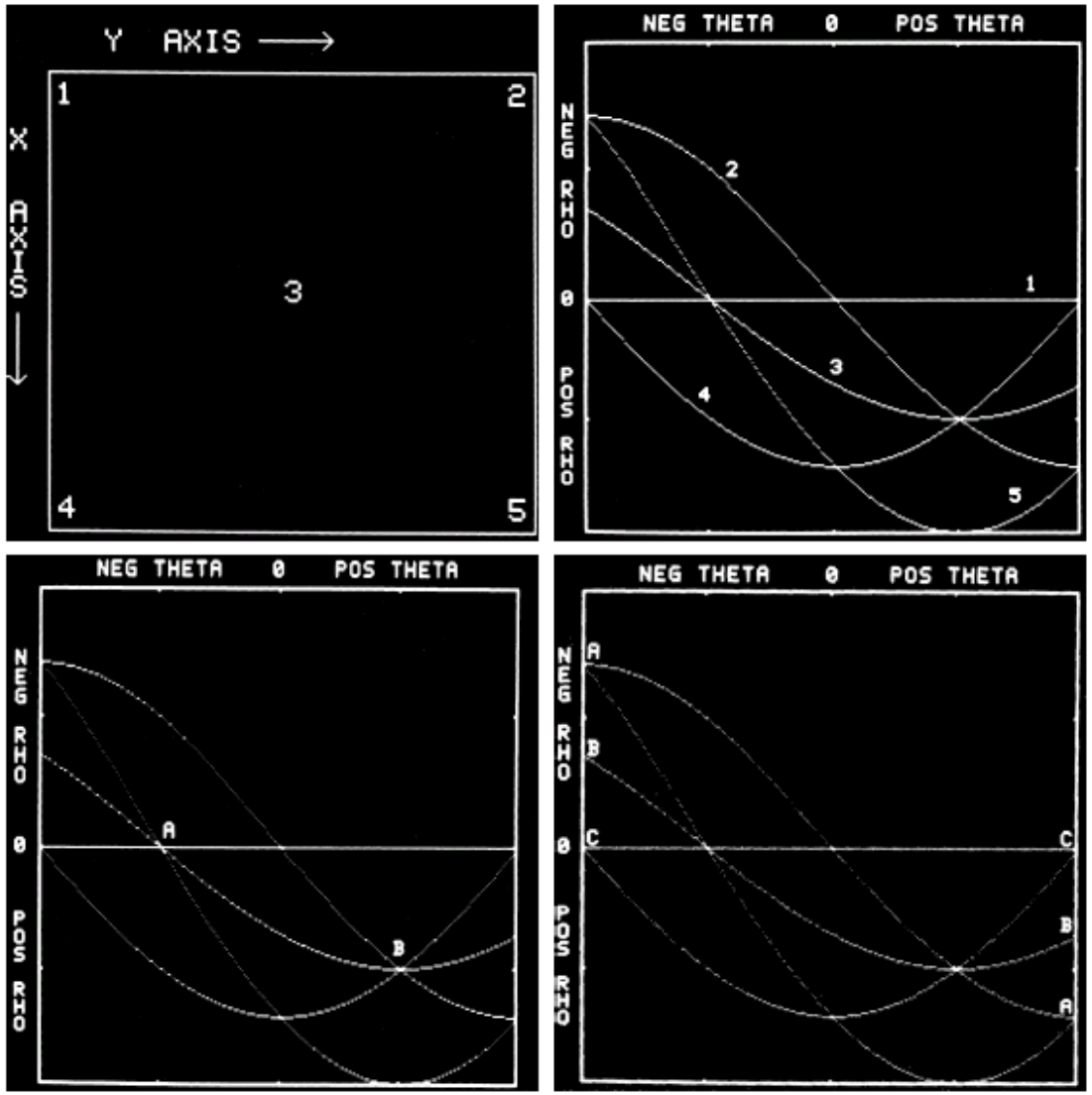
θ = ±90° measured with respect to x-axis

Persamaan x cos θ+ y sin θ = ρ digunakan untuk menggantikan

y = ax + b, supaya menghindari nilai a tak terhingga untuk garis vertikal.

a b
c d

FIGURE 10.20
Illustration of the Hough transform. (Courtesy of Mr. D. R. Cate, Texas Instruments, Inc.)



$$\rho = \pm\sqrt{2}D$$

D: jarak antara 2 titik terjauh

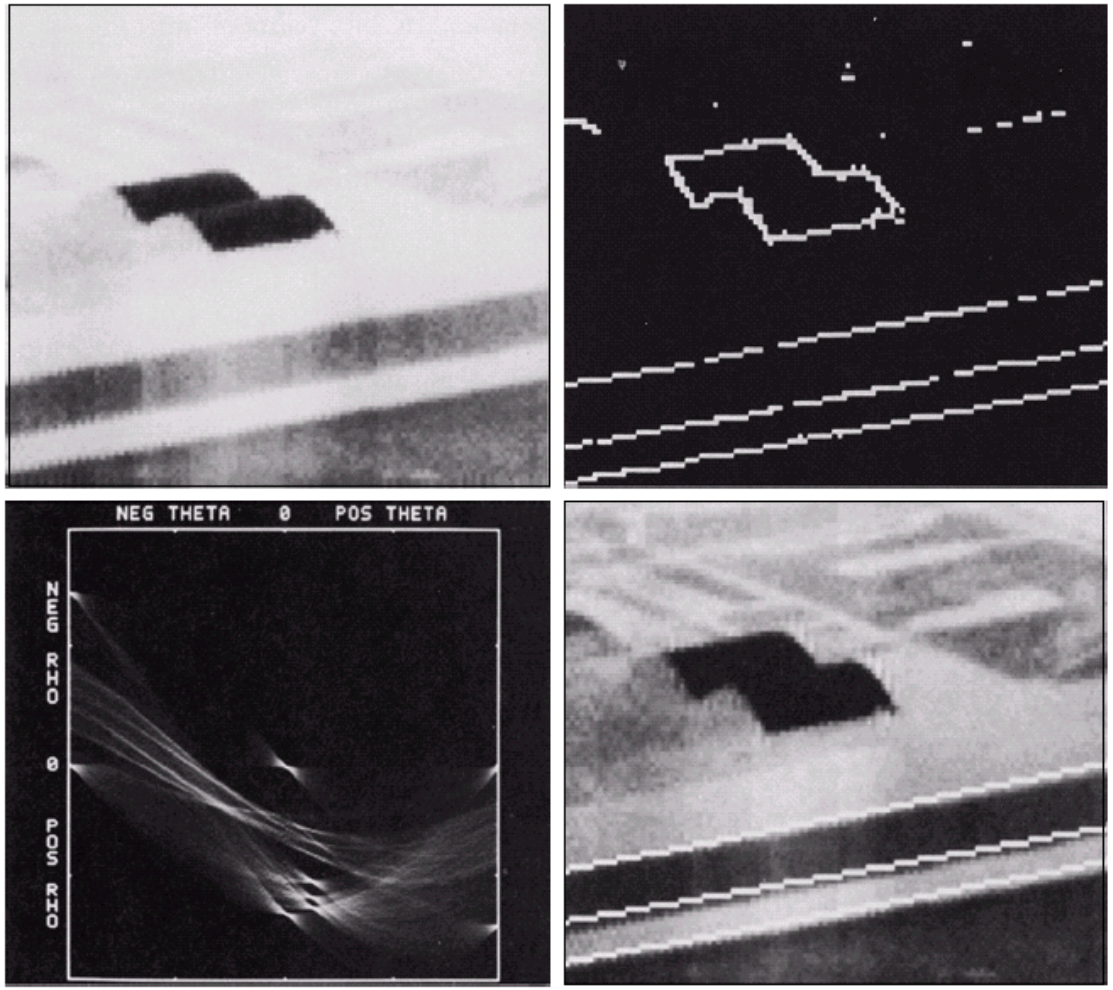
Hough Transformation untuk lingkaran

- Persamaan:

$$(x-c_1)^2 + (y-c_2)^2 = c_3^2$$

- Dengan parameter (c_1, c_2, c_3)
- Akumulator berbentuk kubik

Penyambungan garis berbasis Hough Transform



a b
c d

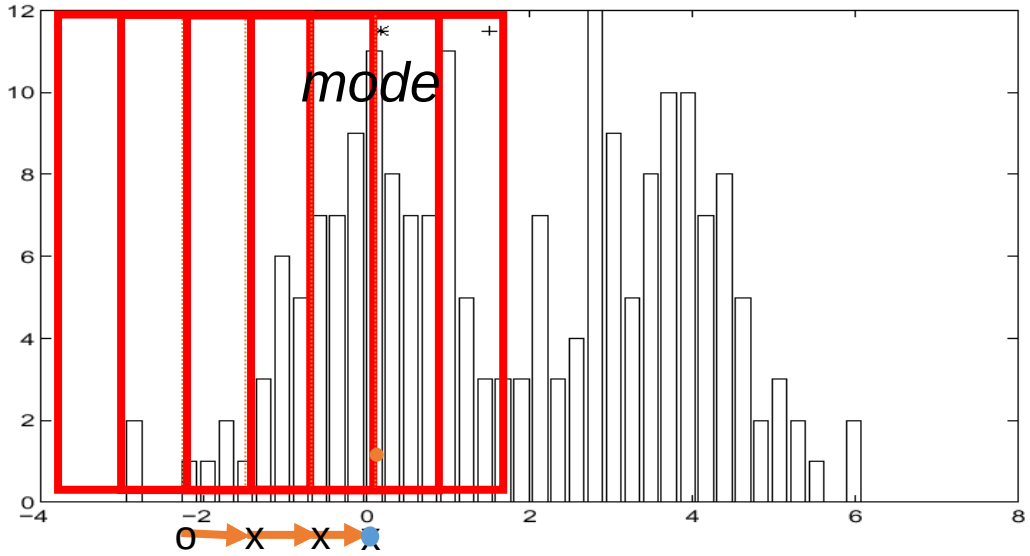
FIGURE 10.21
(a) Infrared image.
(b) Thresholded gradient image.
(c) Hough transform.
(d) Linked pixels.
(Courtesy of Mr. D. R. Cate, Texas Instruments, Inc.)

Segmentasi Citra Lanjutan

- Mean-Shift
- Watershed
- Active Contour

Mean Shift

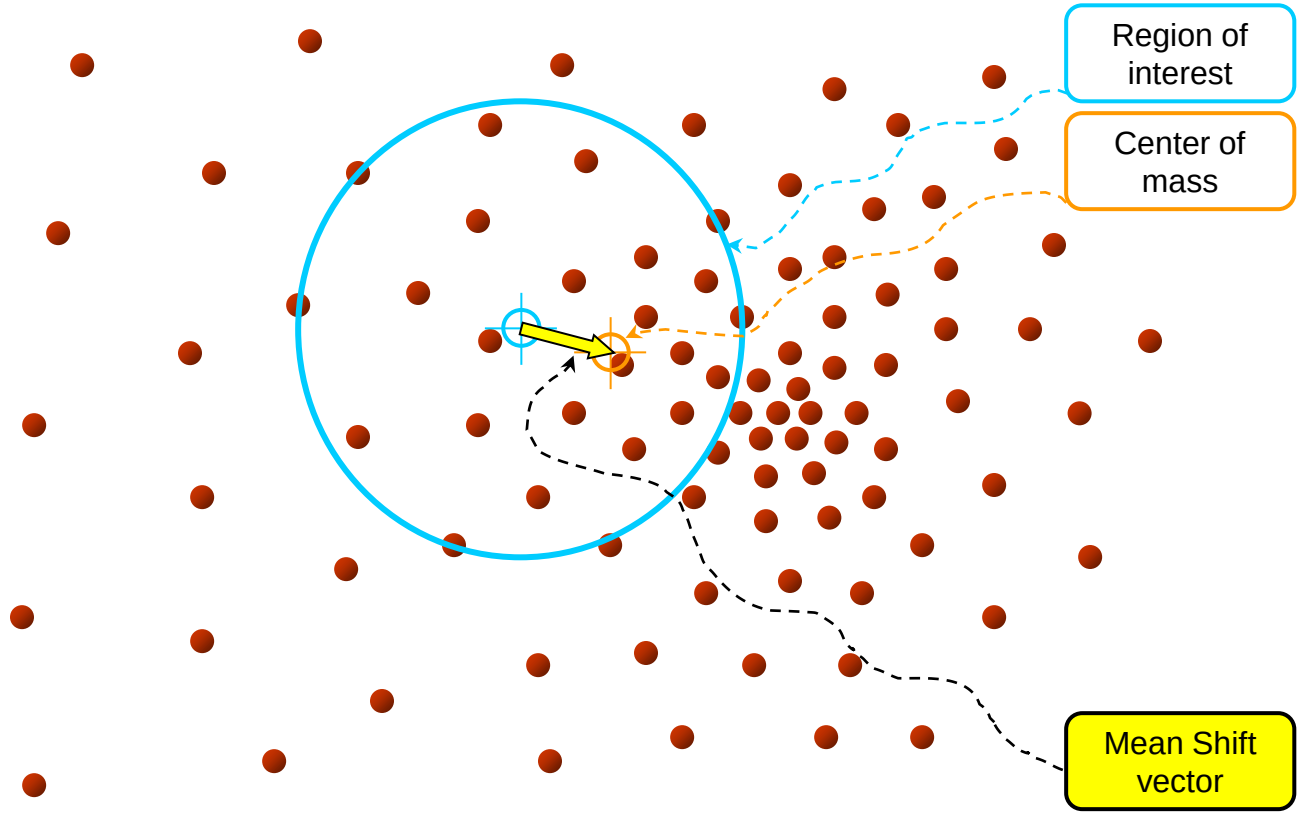
Source: Comaniciu & Meer



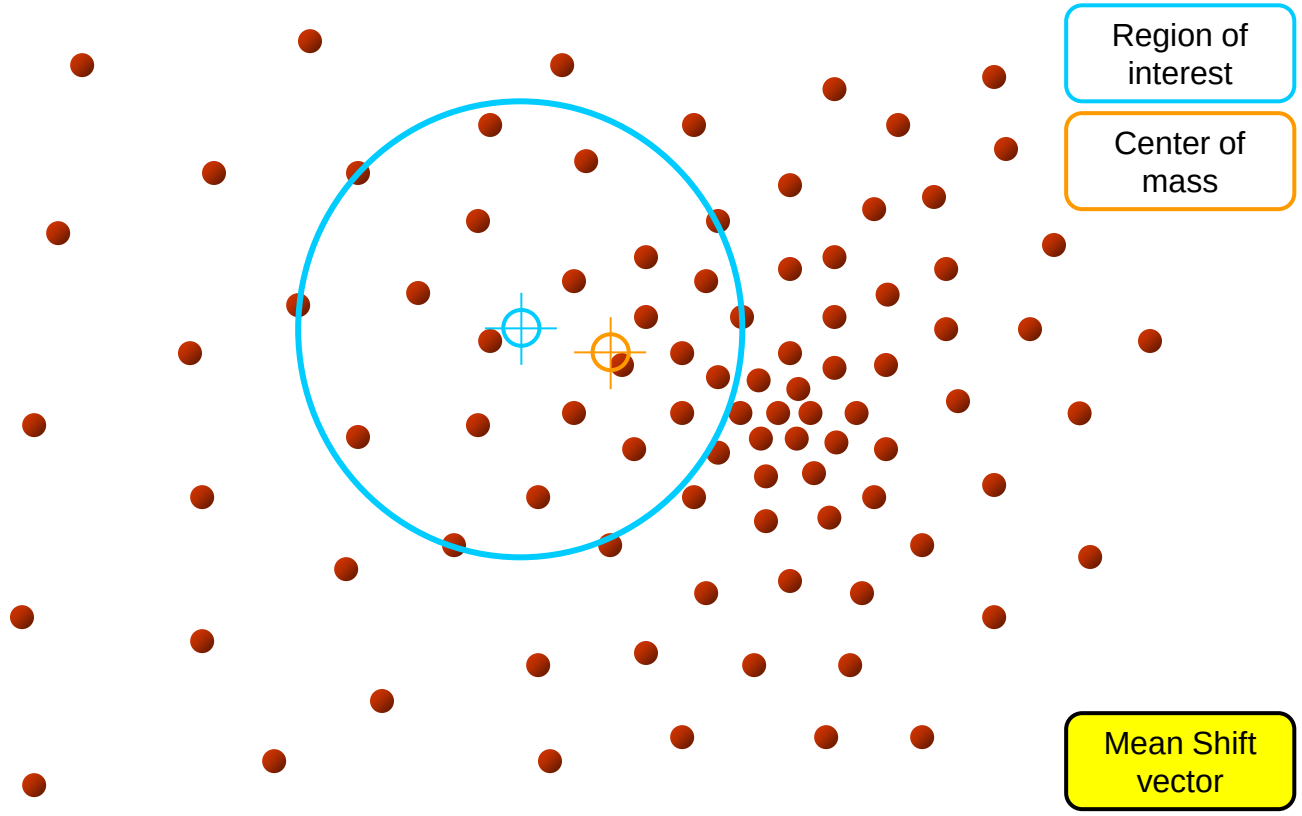
Penelusuran *Mode* secara interatif:

1. Initialisasi *seed* acak
2. Hitung titik pusat/*mean* dari *window* ukuran tetap
3. Geser *window* ke titik pusat
4. Ulangi sampai konvergen

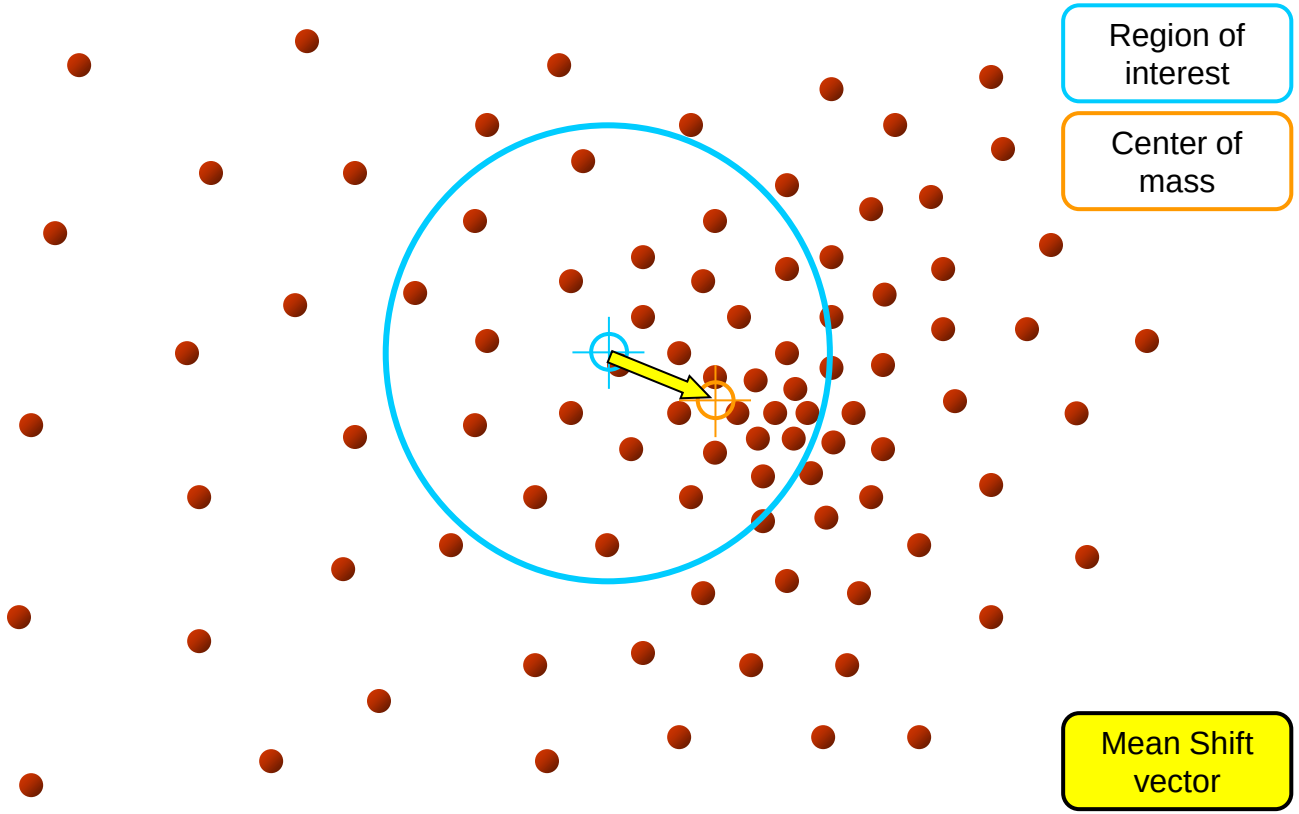
Mean-Shift



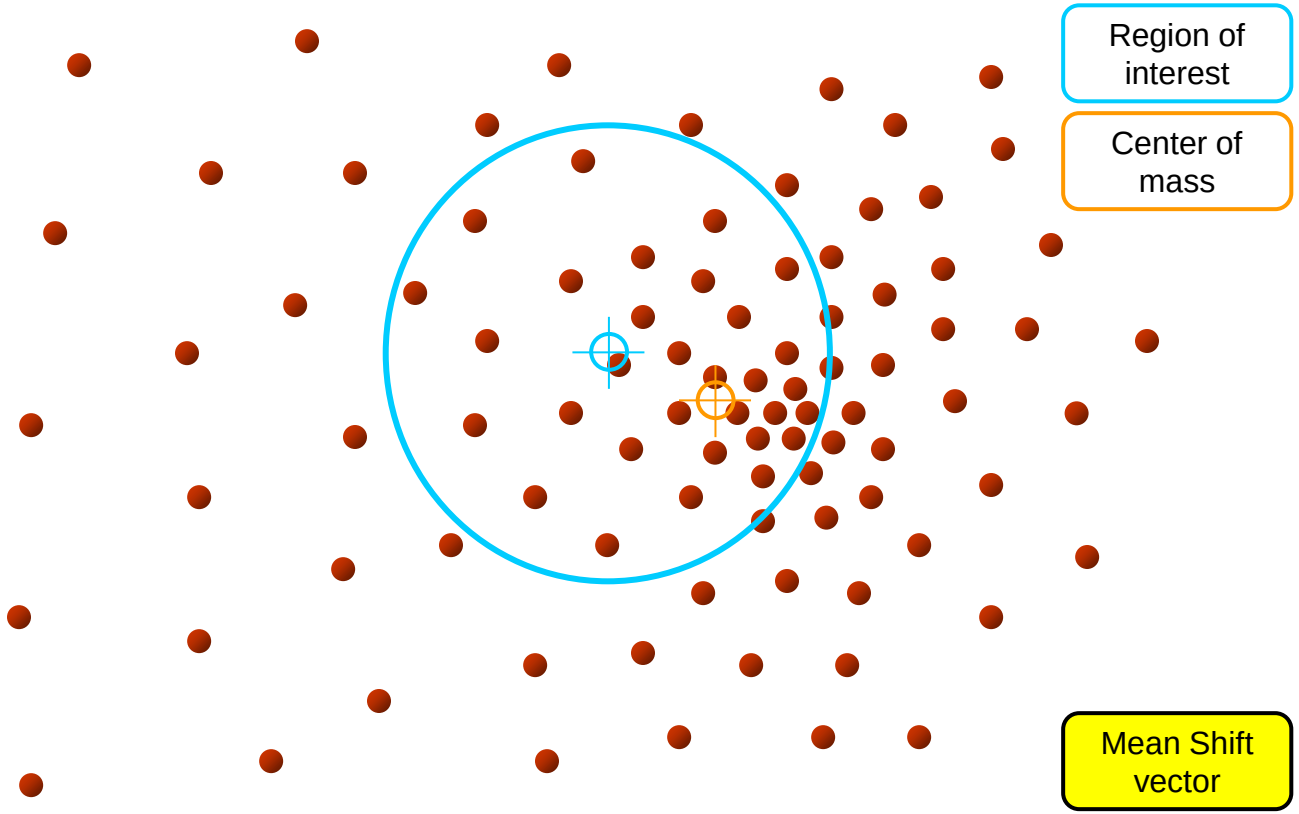
Mean-Shift



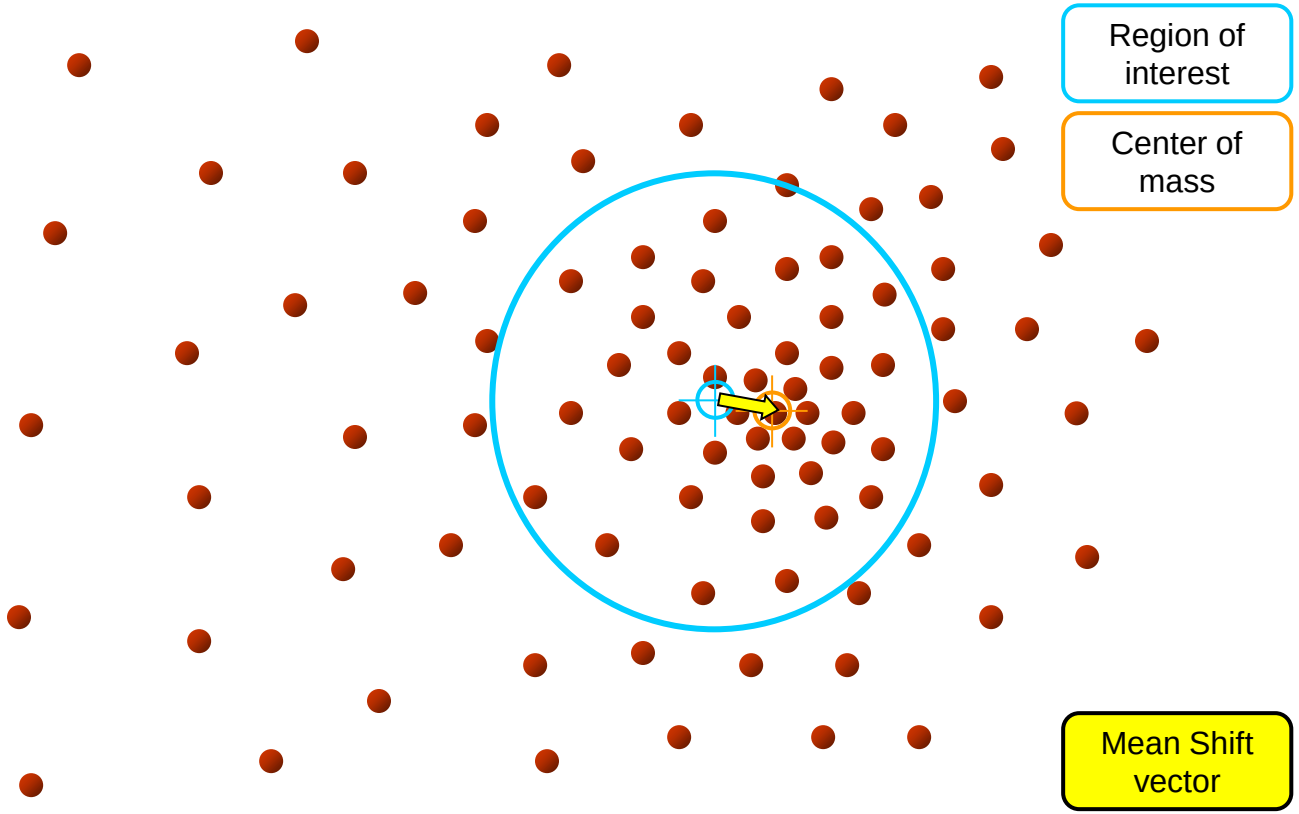
Mean-Shift



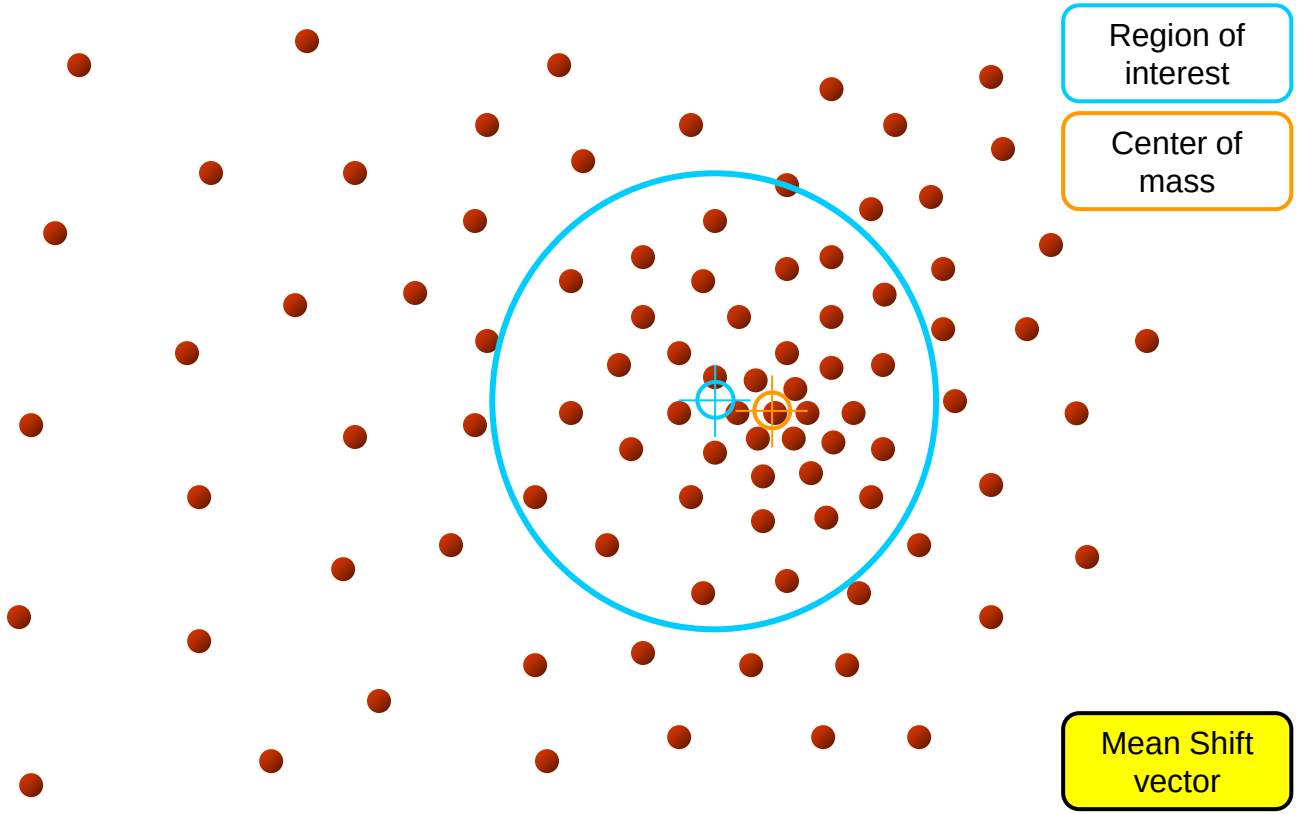
Mean-Shift



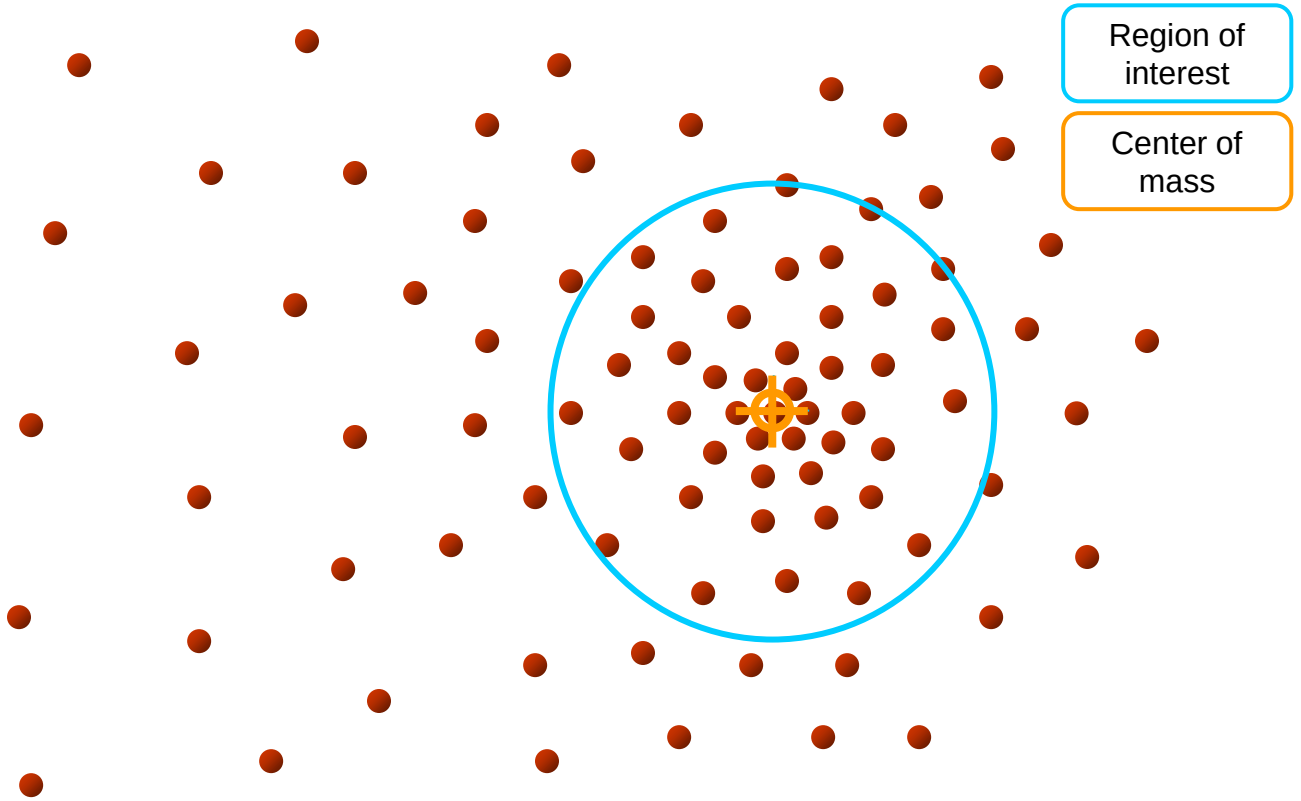
Mean-Shift



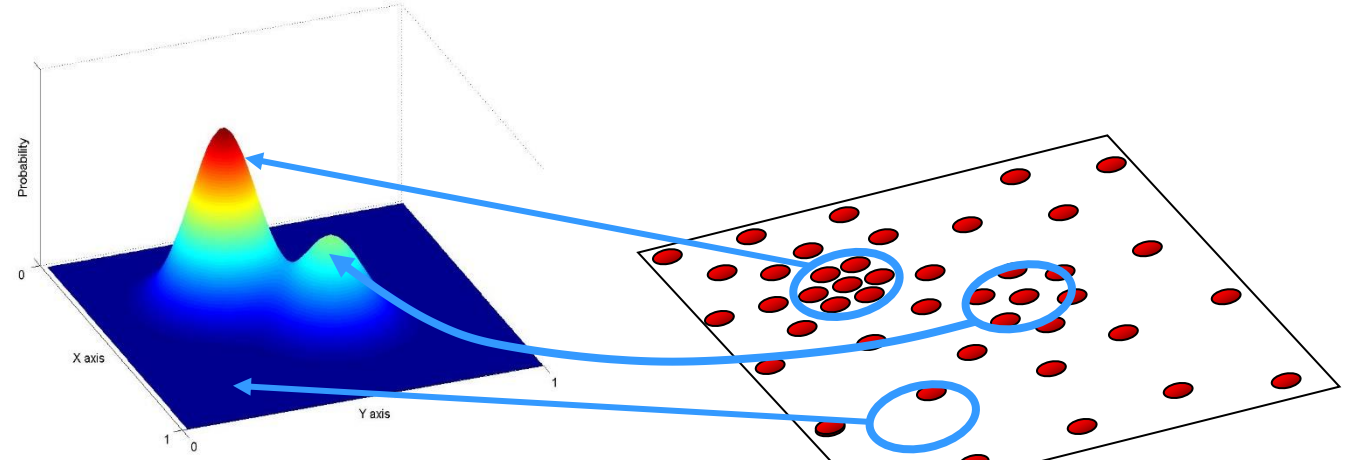
Mean-Shift



Mean-Shift



Non-Parametric Density Estimation



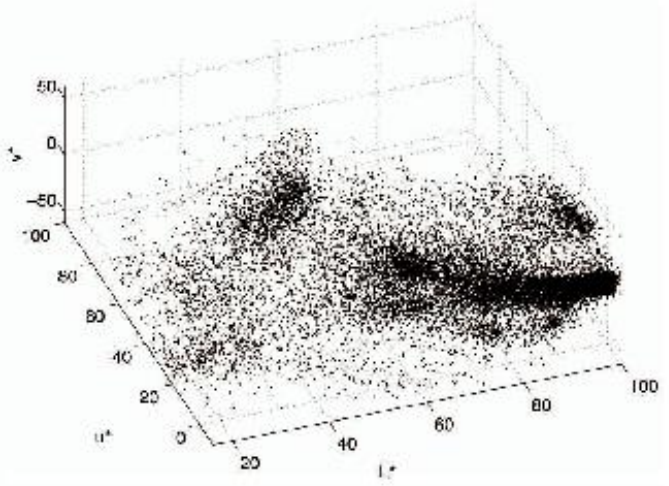
Diasumsikan titik data terdistribusi PDF

Titik data riel

$$\mathrm{PDF}(\mathbf{x}) = \sum_{\mathbf{i}} c_i \cdot e^{-\frac{(\mathbf{x}-\boldsymbol{\mu}_i)^2}{2\sigma_i^2}}$$

Clustering

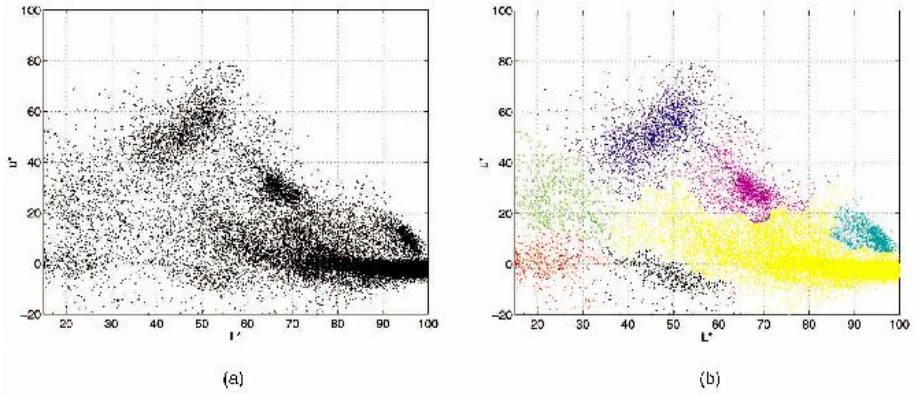




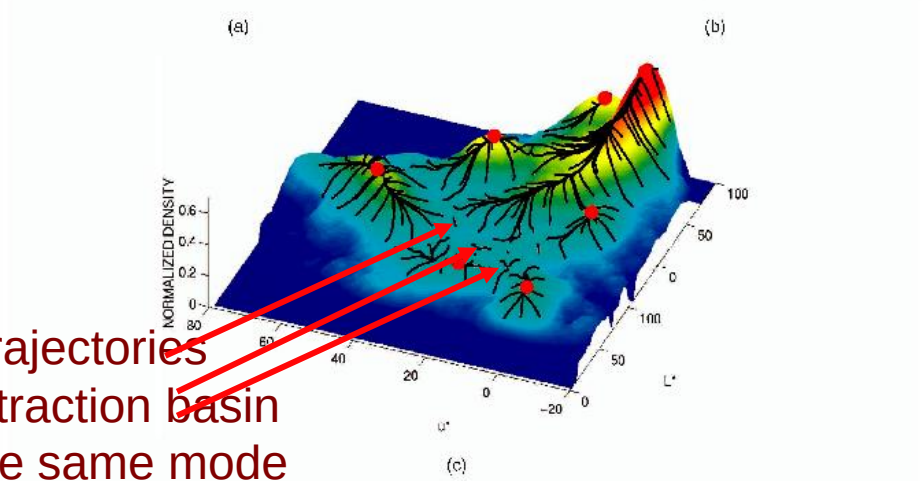
L*u*v space representation

Clustering

2D (L*u) space represent ation

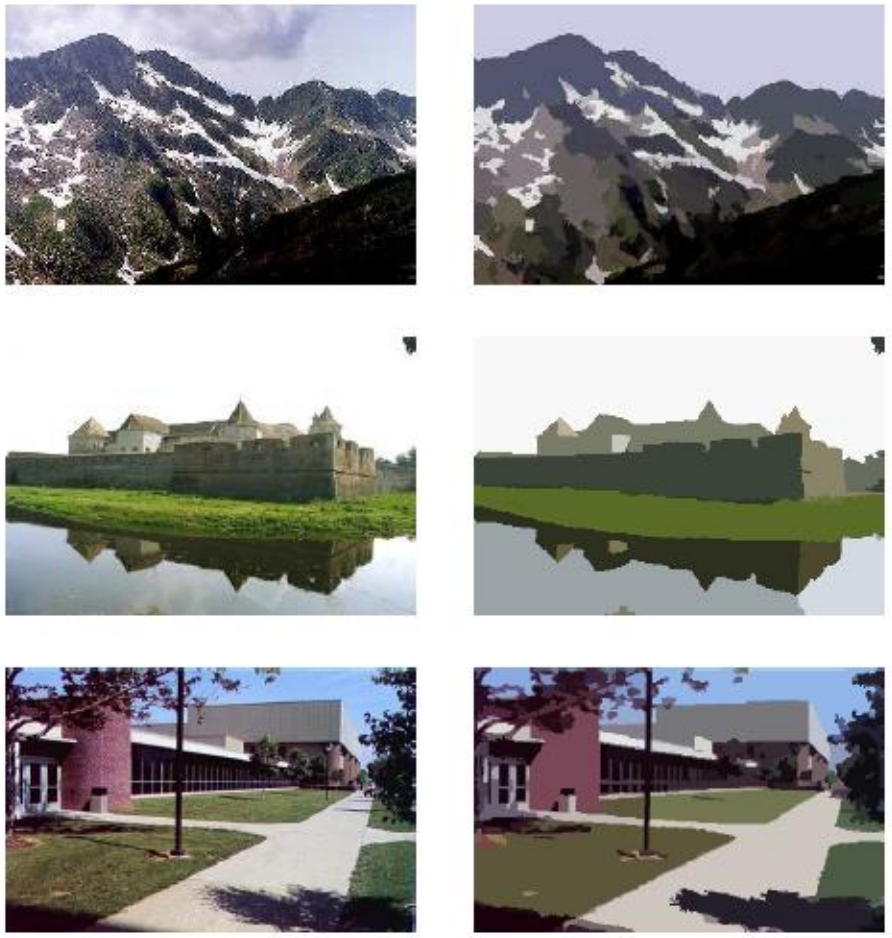


Final clusters



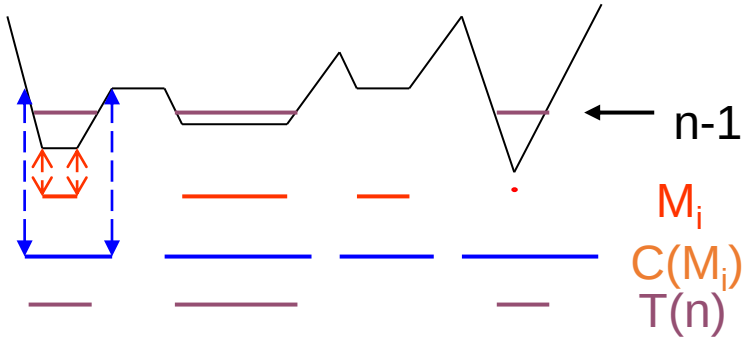
Not all trajectories in the attraction basin reach the same mode

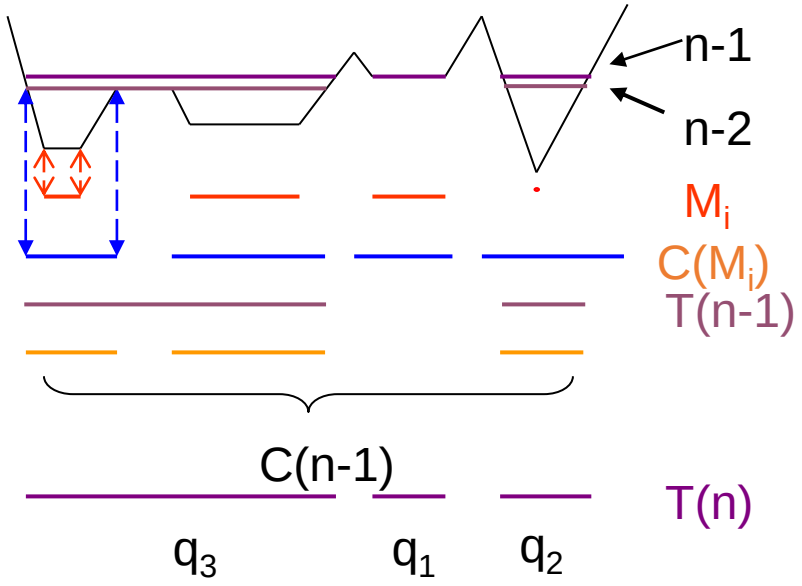
Contoh Hasil Mean-Shift

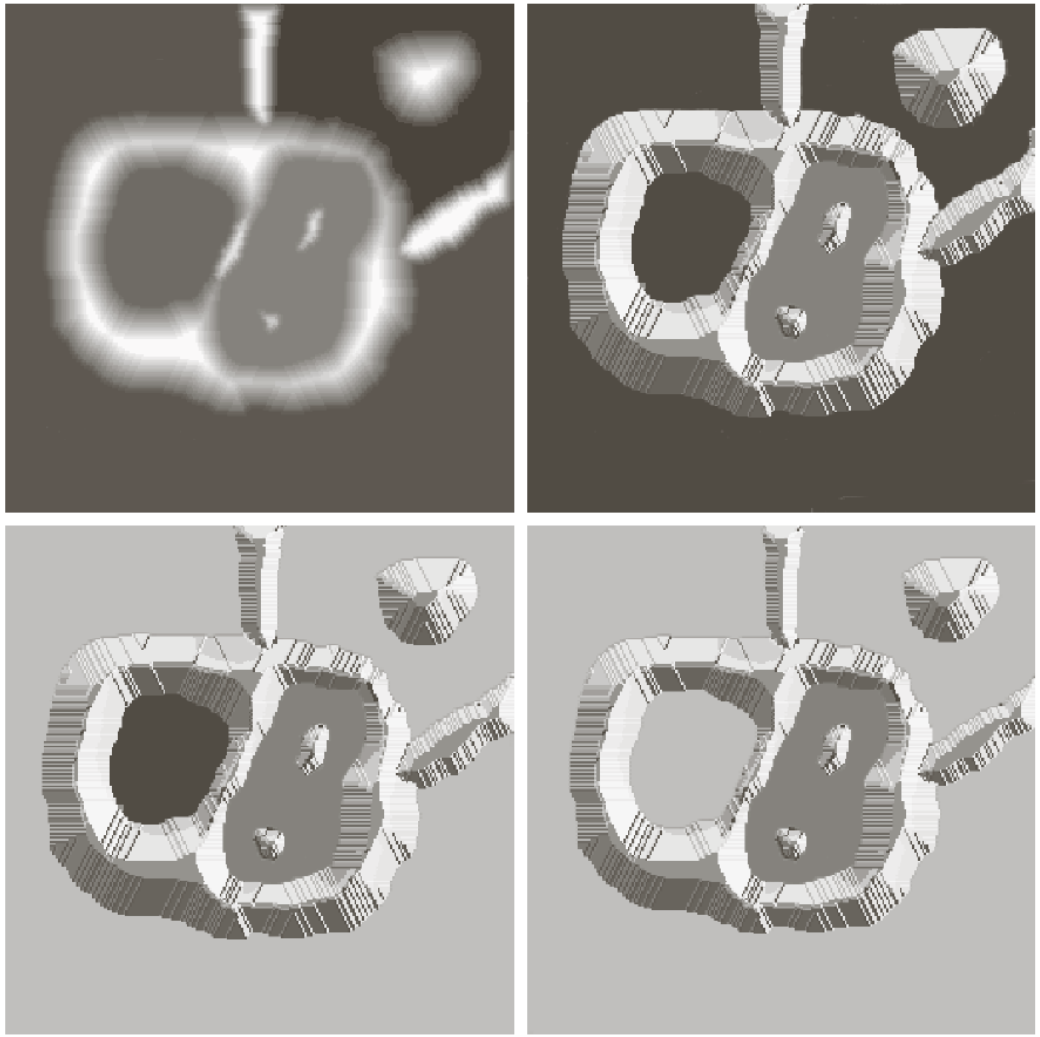


Morphological Watershed

- Berbasis pada dilasi morfologi biner.
- Seringkali digunakan pada citra *gradient* (tiap objek dibedakan dari background oleh tepinya yang menonjol).
- Misal pada contoh, $C(M_i)$ adalah koordinat dari titik-titik dalam *catchment basin* yang terkait dengan *regional minimum* M_i dari g(x,y).
- T[n] adalah koordinat (s,t) dimana g(s,t) < n
- Topografi meluap pada n=max+1.

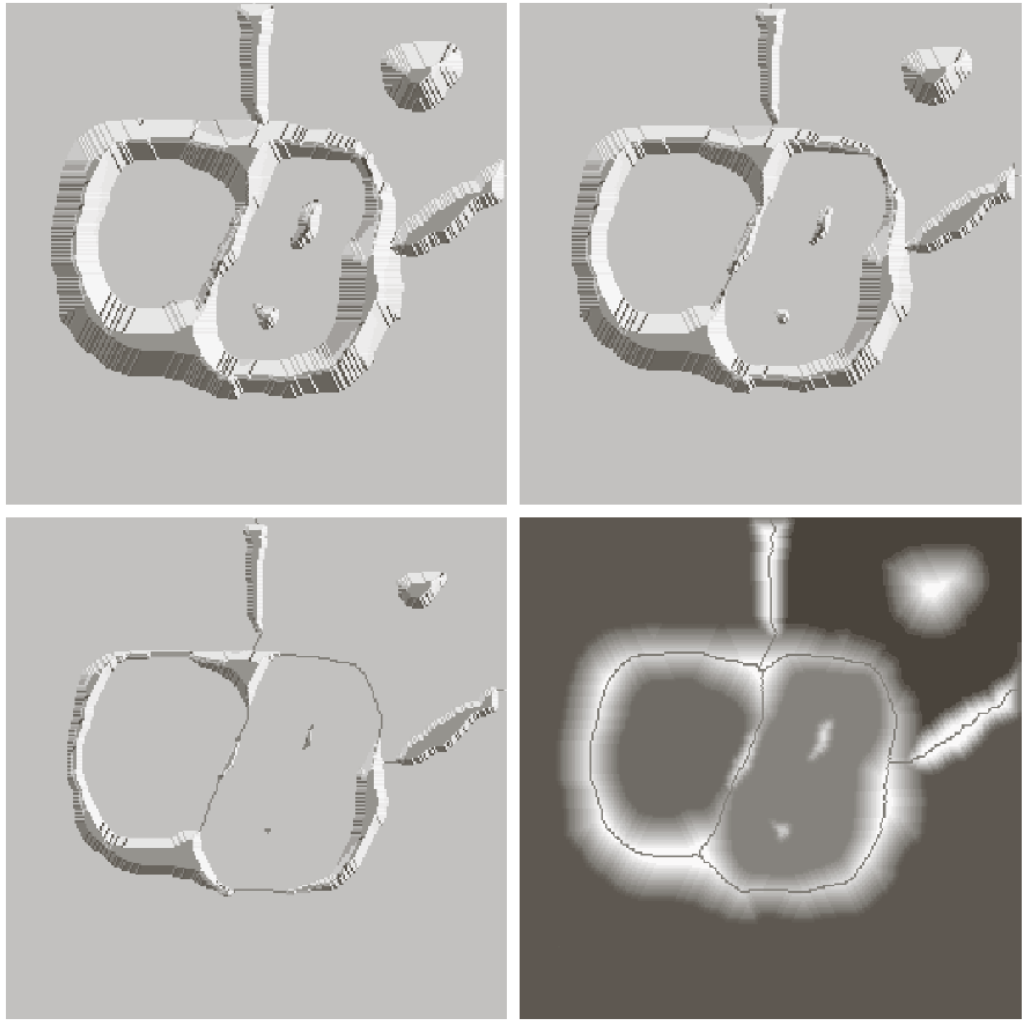






a b
c d

FIGURE 10.54
(a) Original image. (b) Topographic view. (c)–(d) Two stages of flooding.



e f
g h

FIGURE 10.54
(Continued)
(e) Result of further flooding. (f) Beginning of merging of water from two catchment basins (a short dam was built between them). (g) Longer dams. (h) Final watershed (segmentation) lines. (Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

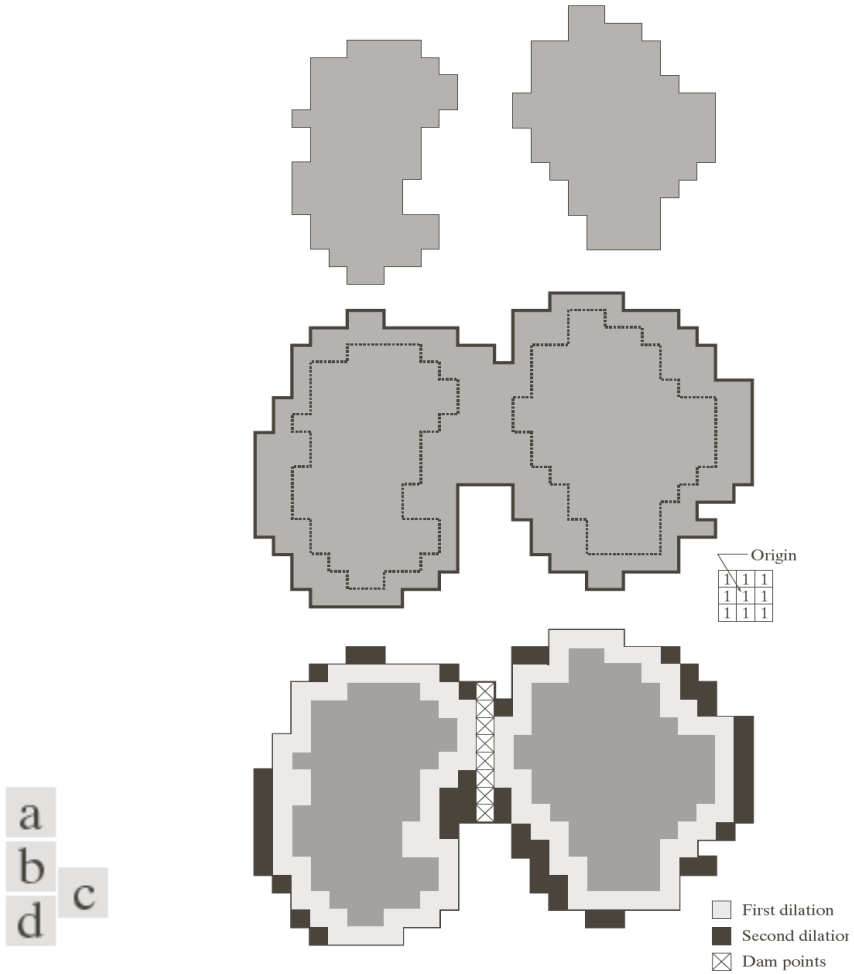
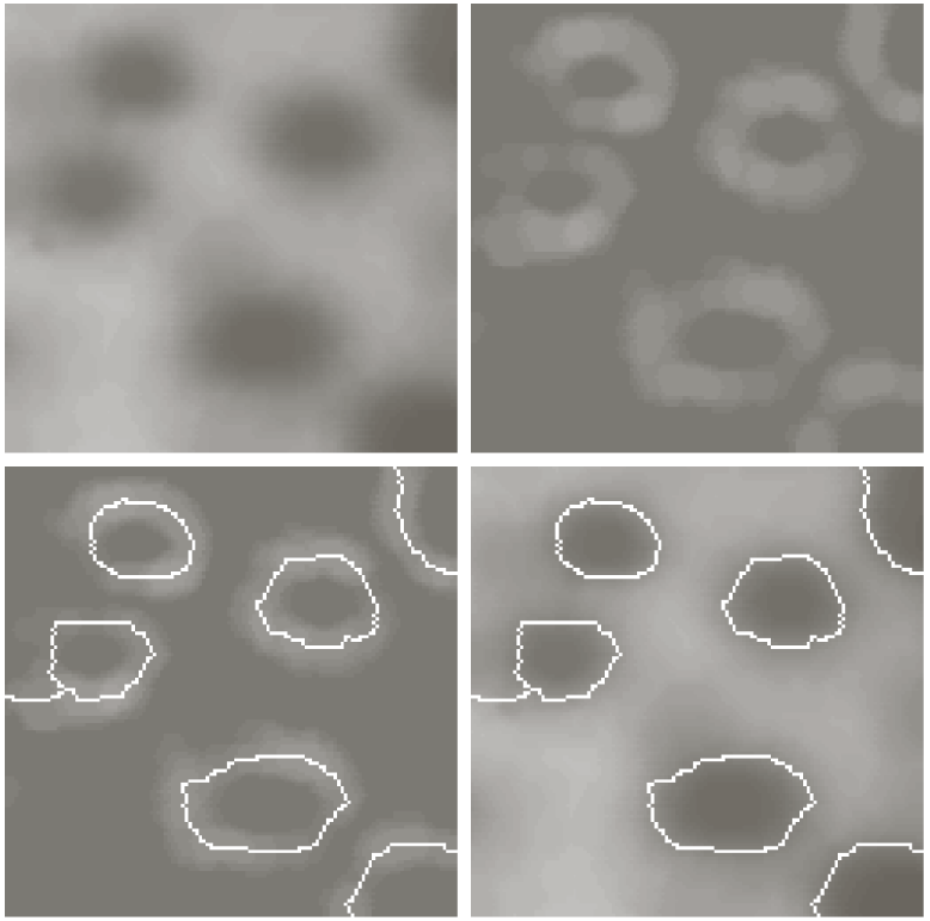


FIGURE 10.55 (a) Two partially flooded catchment basins at stage $n - 1$ of flooding. (b) Flooding at stage n, showing that water has spilled between basins. (c) Structuring element used for dilation. (d) Result of dilation and dam construction.



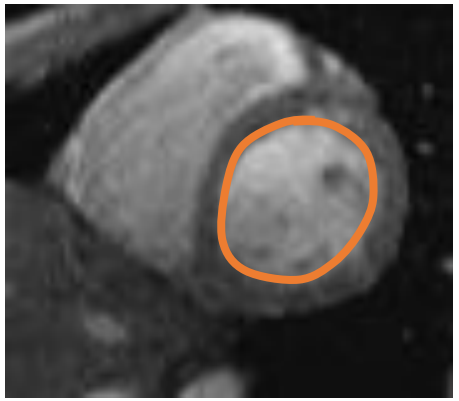
a b
c d

FIGURE 10.56
(a) Image of blobs. (b) Image gradient. (c) Watershed lines. (d) Watershed lines superimposed on original image. (Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

Snakes (active contours):

- Diawali oleh Kass et.al (l987)
- Menyatakan *boundary* objek sebagai kurva parametrik dan minimisasi fungsi enersi terkait dengan kurva
- Diberikan kontur awal dekat objek yang dikehendaki
- Mendapatkan kontur akhir yang sesuai dengan tepi keliling objek





Basic elastic Snakes

Mencari kurva:

$$\mathbf{C} = (\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_{n-1}) = (x_0, y_0, x_1, y_1, ..., x_{n-1}, y_{n-1})$$

Dengan minimisasi fungsi enersi yang terdiri dari enersi internal dan eksternal

$$E = E_{in} + E_{ex}$$

Yang didefinisikan:

$$E = \alpha \cdot \int_0^1 \left| \frac{dv}{ds} \right|^2 ds - \int_0^1 |\nabla I(v(s))|^2 ds$$

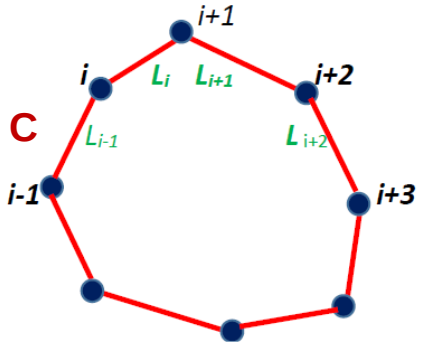
Atau dalam bentuk diskrit:

$$E = \alpha \cdot \sum_{i=0}^{n-1} |v_{i+1} - v_i|^2 - \sum_{i=0}^{n-1} |\nabla I(v_i)|^2$$

Basic elastic Snakes

$$\mathbf{C} = (\mathbf{v}_i \mid 0 \le i < n) = (x_0, y_0, x_1, y_1, \ldots, x_{n-1}, y_{n-1})$$

Enersi internal menjaga kontur halus, enersi eksternal menggerakkan model kearah *boundary*



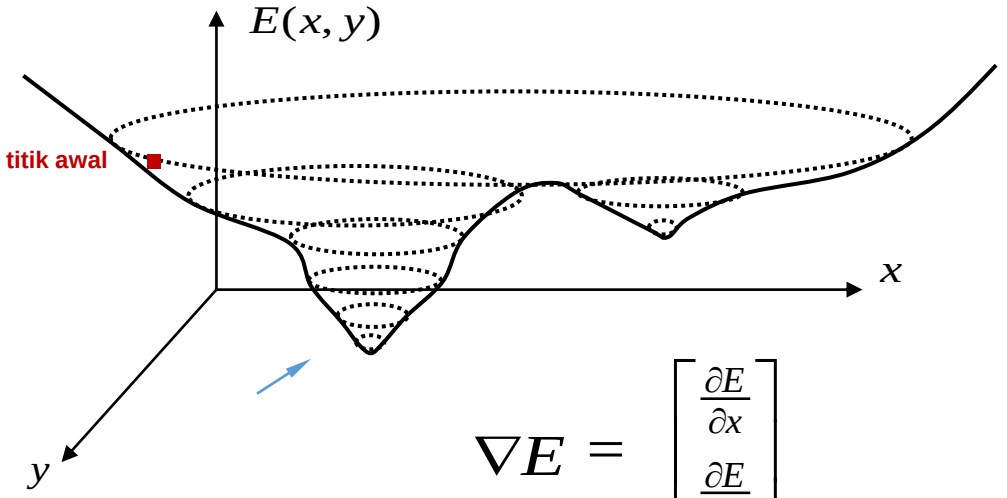
$$E_{in} = \alpha \cdot \sum_{i=0}^{n-1} L_i^2$$

$$= \alpha \cdot \sum_{i=0}^{n-1} (x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2$$

$$E_{ex} = -\sum_{i=0}^{n-1} |\nabla I(x_i, y_i)|^2$$

$$= -\sum_{i=0}^{n-1} |I_x(x_i, y_i)|^2 + |I_y(x_i, y_i)|^2$$

Gambar minimisasi fungsi 2 variabel dengan Gradient Descent



$$\nabla E = \begin{bmatrix} \frac{\partial E}{\partial x} \\ \frac{\partial E}{\partial y} \end{bmatrix}$$

Gradient negatif memberikan arah *steepest descent* kearah fungsi E dengan nilai lebih rendah

Masalah yang dapat timbul

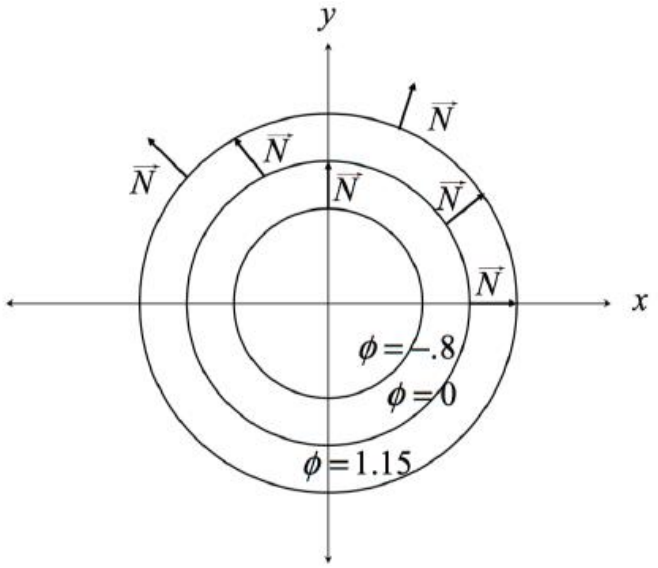
- Inisial sangat menentukan
- Hasil tergantung pada jumlah dan jarak titik control
- Hasil tidak sesuai dengan perubahan topologi objek
- Dapat terjadi perpotongan dalam kurva:



- Batasan pada active contour yang berbasis pada kurva parametric adalah problem yang terjadi pada perubahan topologi kurva, hal ini diatasi dengan *curve reparameterization*.
- Alternatif solusi dengan menggunakan level set (LS) yang menelusuri jejak objek dengan memodifikasi fungsi embedding yang mendasari

Contoh: Kurva sebagai Level Set dari suatu Fungsi

$$\phi(x, y) = x^2 + y^2 - 1 = 0$$



$$\nabla\phi = (\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y})$$

$$\vec{N} = \frac{\nabla\phi}{|\nabla\phi|}$$

$$\frac{\partial\phi}{\partial x} \approx \frac{\phi_{i+1} - \phi_i}{\Delta x}$$

Chan-Vese functional

Dengan Φ merupakan fungsi level set, maka *Chan-Vese functional* adalah,

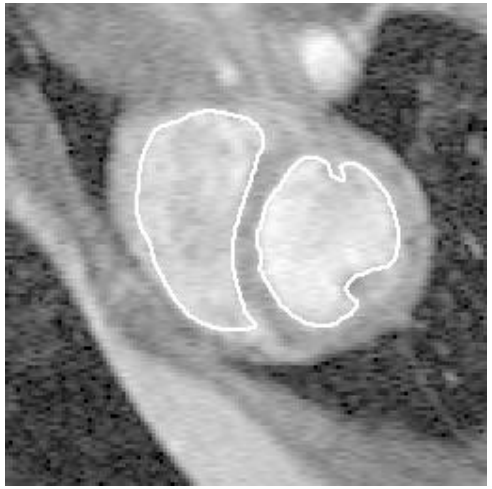
$$F(\phi, c_1, c_2) = \mu \int_\Omega \delta(\phi)|\nabla\phi| + \nu \int_\Omega H(\phi)dxdy$$
$$+ \lambda_1 \int_\Omega |u_0 - c_1|^2 H(\phi)dxdy + \lambda_2 \int_\Omega |u_0 - c_2|^2 (1 - H(\phi))dxdy$$

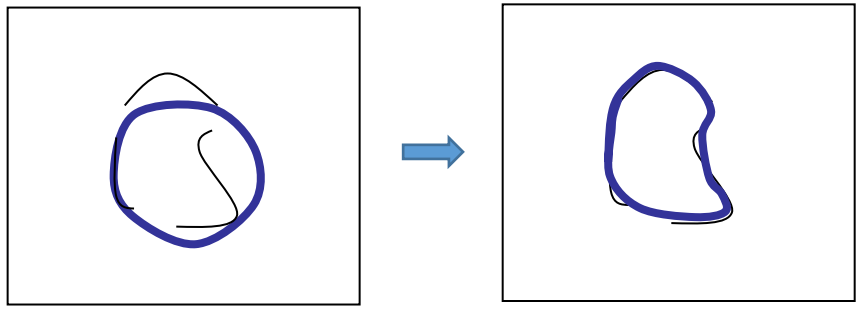
Dimana u_0 adalah citra yang diberikan, c_1 dan c_2 adalah harga rata-rata u_0 didalam dan diluar kurva C (*evolving curve*), H adalah fungsi Heaviside.

Ref: An Active Contour Model without Edges (Chan & Vese)

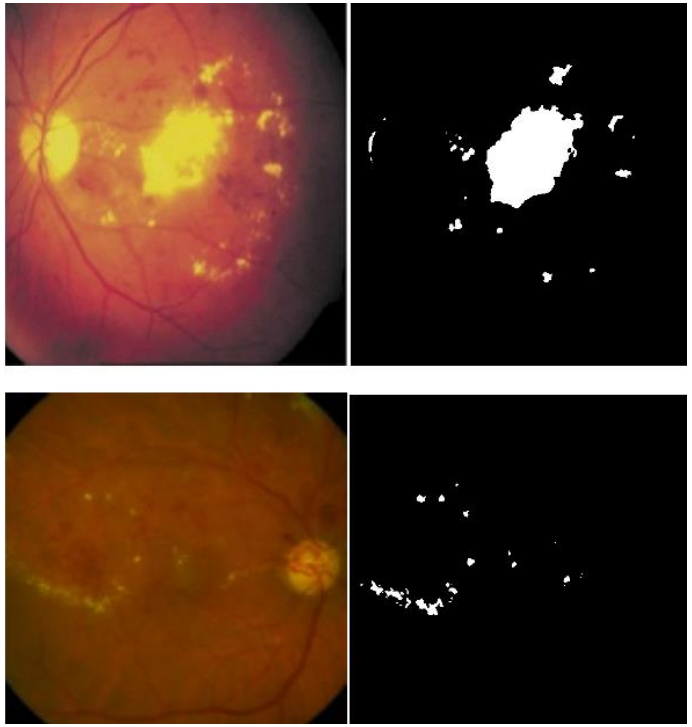
Contoh:

Source:Visual Dynamics Group (University of Oxford)





Contoh Segmentasi:
SEGMENTATION OF HARD EXUDATES



Menghitung secara iteratif membership dan centroids FCM yang dimodifikasi :

$$\mu k(xi, yi) = \frac{(wi - vk)^{-2/m-1}}{\sum_{j=0}^{c-1}(wi - vj)^{-2/m-1}}$$

$$vk = \frac{\sum_{i=0}^{q-1} \mu k\,(xi,yi)^m wi}{\sum_{i=0}^{q-1} \mu k\,(xi,yi)^m}$$

Menghitung akurasi ,
A=ground truth, R=hasil:

$$P(R;A) = \frac{|A \cap R|}{|A \cup R|} \times 100\%,$$

H. Tjandrasa, A. Y. Wijaya, I. Arieshanti, N. D. Salyasari, "Segmentation of Hard Exudates in Retinal Fundus Images Using Fuzzy C-Means Clustering with Spatial Correlation", The Proceedings of The 7th ICTS, 2013.

Contoh Preprocessing:
Optic Nerve Head Segmentation

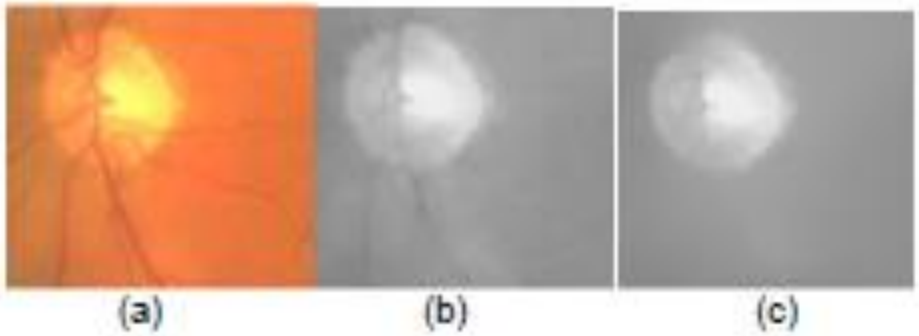
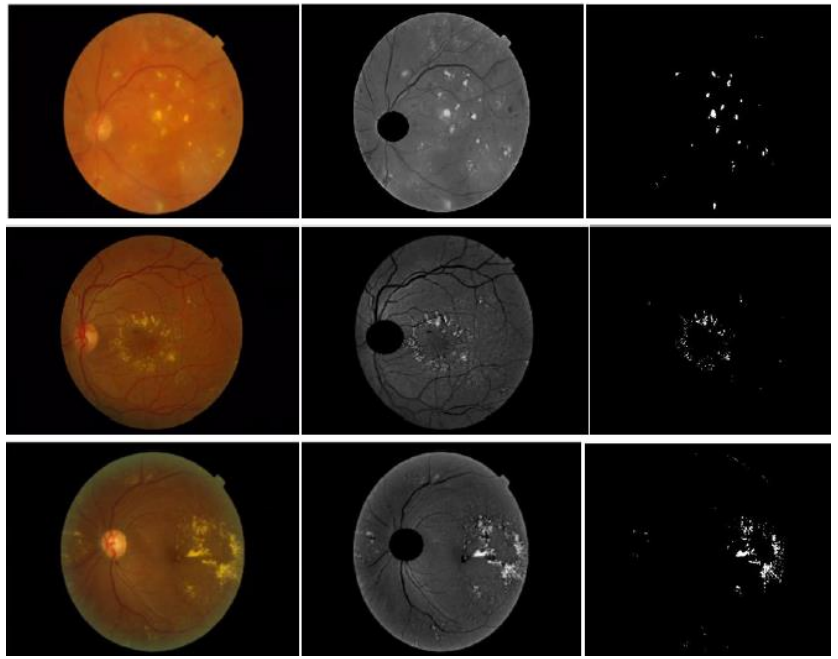


Figure 2. Vessel removal result: (a) input image; (b) after Homomorphic filtering, (c) after vessel removal

H. Tjandrasa, A. Wijayanti, N. Suciati, "Optic Nerve Head Segmentation Using Hough Transform and Active Contours", TELKOMNIKA, September 2012.

Contoh Klasifikasi:

Classification of Non-Proliferative Diabetic Retinopathy Based on Segmented Exudates using K-Means Clustering



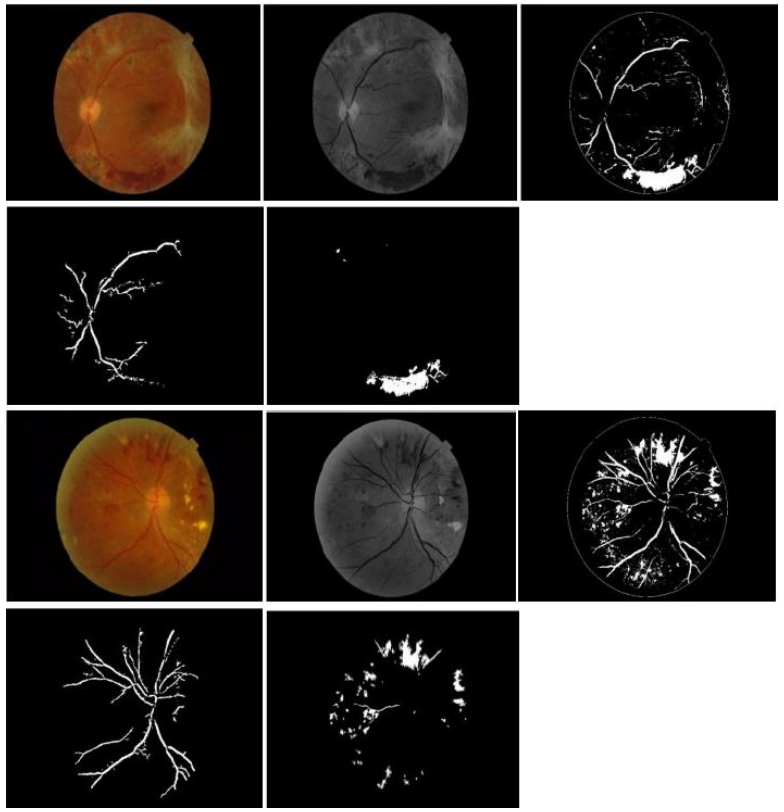
No.	Classifier	Accuracy
1	SVM	89,29%
2	Multilayer Perceptron	91,07%
3	RBF Network	85,71%

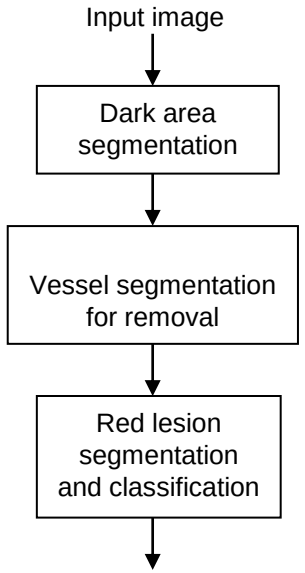
Handayani Tjandrasa, Isye Arieshanti, Radityo Anggoro, “Classification of Non-Proliferative Diabetic Retinopathy Based on Segmented Exudates using K-Means Clustering”, IJIGSP, 2015

Contoh Klasifikasi:

Classification of Non-Proliferative Diabetic Retinopathy Based on Features Extracted from Red Lesion Segmentation

(H. Tjandrasa et al.)






THANK YOU